



**Verified Carbon
Standard**

PADANG TIKAR LANDSCAPE



Document Prepared by South Pole

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Padang Tikar Landscape project is located in the Padang Tikar Landscape of the Batu Ampar subdistrict, Kubu Raya district, West Kalimantan province, Indonesia. The project consists of 10 'Village Forests', covering 76,370 hectares (ha), 58,672.7 ha of which is eligible for the reducing emissions from deforestation and forest degradation (REDD) project. The Village Forest of Padang Tikar is the largest 'Village Forest' permit in Indonesia and is divided into three types of forest: protected forest (69,299 ha), limited production forest (1,985 ha), and convertible production forest (5,086 ha). The Padang Tikar Landscape project is a microcosm of the country as it has comprehensive forest ecosystems, ranging from inland forests to swamp forests (including mangrove forests and peatlands).

Before the project began, illegal logging in mangroves and agricultural areas was the most widespread land use in the project area (PA), causing average annual forest losses of around 529 ha per year.¹ This activity contributed not only to growing insecurity in the region, but also to a lack of education and employment opportunities. These conditions created a reinforcing loop, where locals felt that exploiting surrounding ecosystems through logging and unsustainable agriculture was the only way to advance economically.

The overall objective of this project is to secure and protect the Village Forest of Padang Tikar and restore and improve ecosystem services and habitat through conservation activities. The project will also bring opportunities for community development and improve the lives and livelihoods of locals. Community members have been consulted extensively through a participatory approach and have identified several activities that can contribute to this outcome.

Indonesia's Ministry of Environment and Forestry (MoEF) granted a 'Village Forest' license to Padang Tikar on 14 February 2017.² The Village Forest of Padang Tikar falls within the Padang Tikar Landscape, which contains 10 villages: Tanjung Harapan, Ambarawa, Sungai Besar, Sungai Jawi, Tasik Malaya, Padang Tikar I, Medan Mas, Nipah Panjang, Teluk Nibung, and Batu Ampar. The primary objective is the reduction of greenhouse gas (GHG) emissions through increasing forest sustainability, enhancing community welfare, reducing pressure on land availability by focusing on creating a fair and sustainable system in managing coastal areas, and conducting community engagement and training to enhance the capacity of villages in the Village Forest of Padang Tikar.

¹ According to historical analysis.

² The supporting documents are available in the folder Section 1/Village Forest Permit.

In addition, it is expected that this project will provide community benefits by strengthening local governance, enhancing sustainable community management practices around Village Forests, and increasing the perception of the value of forest resources. This project will also lead to the economic empowerment of community groups and an increase in employment opportunities by:

- 1) increasing the ability of the forest patrol to protect and safeguard forests and biodiversity;
- 2) restoring critical mangrove and peat ecosystems;
- 3) strengthening the community's sustainable livelihood business model; and
- 4) increasing community capacity through training, comparative studies, and internships.

Following implementation of the project activities, the project is expected to generate estimated average annual GHG emission reductions of 590,557 metric tons of carbon dioxide equivalent (tCO₂e) and total reductions of 5,905,572 tCO₂e for the initial ten-year baseline period.

1.2 Sectoral Scope and Project Type

This project falls under scope 14 of the Verified Carbon Standard (VCS) 'Agriculture, Forestry and Other Land Use' (AFOLU) guidelines. The project type is mixed, including both a REDD component, in the category of Avoided Unplanned Deforestation and Degradation (AUDD), and a Wetland Restoration and Conservation (WRC) component. It is a non-grouped project that seeks to reduce deforestation, protect mangrove areas in the Padang Tikar Landscape, and conserve organic carbon stocks in peatlands.

1.3 Project Eligibility

According to Appendix 1 'Eligible AFOLU Project Categories under VCS Standard v4.2', REDD activities are those that reduce net GHG emissions by reducing the deforestation and/or degradation of forests.³ Deforestation is the direct, human-induced conversion of forest land to non-forest land.

The PA meets an international definition of 'forest' and was qualified as forest for a minimum of 10 years before the project start date. The definition of 'forest' may include mature forests, secondary forests, and degraded forests. Under the VCS Program, 'secondary forests' are forests considered as having been cleared, having recovered naturally, being at least 10 years old, and meeting the lower bound of the forest threshold parameters at the start of the project.

³ Available at https://verra.org/wp-content/uploads/2022/02/VCS-Standard_v4.2.pdf.

The main effect of this project activity is reducing carbon emissions by preventing the conversion of forest with high carbon stocks (HCSs) to non-forest with lower carbon stocks. In addition, this activity can also help avoid the conversion of forest areas to cropland and assist with the introduction of agricultural activities which can reduce the methane (CH₄) emissions associated with the burning of biomass to clear land. The deforestation configuration applicable to this project, as defined by the methodology, is 'frontier', described as the result of the expansion of small-scale agriculture activities (e.g., subsistence farming) and illegal logging.⁴

1.4 Project Design

- ☒ The project includes a single location or installation only
- ☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☐ The project is a grouped project

The includes a single location only. The project has been designed as an AUDD, applying the VM0007 REDD+ Methodology Framework (REDD+MF), v1.6, dated September 8, 2020.

Eligibility Criteria

This is not applicable as the project is a non-grouped project.

1.5 Project Proponent

Organization name	Ambarawa village
Contact person	M Bakri M. Dona Darwis
Title	Head of village Head of LPHD Head of BPD
Address	Jl. Raya Ambarawa, Kecamatan Batu Ampar, Kabupaten Kubu Raya Provinsi Kalimantan Barat, Indonesia, 78385

⁴ 'Frontier configurations' are described as any landscape in which none of the forest in the project area has a current and direct physical connection with areas that have been deforested anthropogenically.

Telephone	+62 816-4968-6237
Email	-

Organization name	Batu Ampar village
Contact person	Herry Prawiranto Akhmad. S Syarif Rusli
Title	Head of village Head of LPHD Head of BPD
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Telephone	+62 821-5385-3882
Email	-

Organization name	Medan Mas village
Contact person	Dharma Wira Rusdi Eka Rahadi
Title	Head of village Head of LPHD Head of BPD
Address	Jl. Arahman, Dusun Medan Deli, RT 002/RW 001, Kecamatan Batu Ampar, Kabupaten Kubu Raya, Provinsi Kalimantan Barat, Indonesia, 78385
Telephone	-
Email	-

Organization name	Nipah Panjang village
Contact person	Mochtar D Toni Shalihin
Title	Head of village Head of LPHD Head of BPD
Address	Jl. Raya Nipah Panjang RT 003/ RW 001, Kecamatan Batu Ampar, Kabupaten Kubu Raya Provinsi Kalimantan Barat, Indonesia, 78385
Telephone	-
Email	-

Organization name	Padang Tikar Satu village
Contact person	Mujahidin Iwan Kurniawan M. Syahid
Title	Head of village Head of LPHD Head of BPD
Address	JL. Raya Patimura RT 001/ RW 002, Kecamatan Batu Ampar, Kabupaten Kubu Raya, Provinsi Kalimantan Barat, Indonesia, 78385
Telephone	-
Email	-

Organization name	Sungai Besar village
Contact person	Sumar Mulyadi M. Arif

Title	Head of village Head of LPHD Head of BPD
Address	Jl. Raya Sungai Besar RT 002/RW 001, Kecamatan Batu Ampar Kabupaten Kubu Raya, Provinsi Kalimantan Barat, Indonesia, 78385
Telephone	-
Email	-

Organization name	Sungai Jawi village
Contact person	Effendi Abu Bakar Sapei
Title	Head of village Head of LPHD Head of BPD
Address	Jl. Raya Sungai Jawi RT 003/RW 001, Kecamatan Batu Ampar, Kabupaten Kubu Raya, Provinsi Kalimantan Barat, Indonesia, 78385
Telephone	-
Email	-

Organization name	Tanjung Harapan village
Contact person	Sahron Joni Sahda
Title	Head of village Head of LPHD Head of BPD
Address	Jl. Raya Tanjung Harapan, RT 003/ RW 001, Kecamatan Batu Ampar, Kabupaten Kubu Raya, Provinsi Kalimantan Barat, Indonesia, 78385

Telephone	-
Email	-

Organization name	Tasik Malaya village
Contact person	Hendratullah Yakub Uni Harnipul
Title	Head of village Head of LPHD Head of BPD
Address	Jl. Raya Tasik Malaya, RT 002/ RW 001, Kecamatan Batu Ampar, Kabupaten Kubu Raya, Provinsi Kalimantan Barat, 78385
Telephone	-
Email	-

Organization name	Teluk Nibung village
Contact person	Gunawan Mahyudin Bohari
Title	Head of village Head of LPHD Head of BPD
Address	Desa Teluk Nibung RT 005/RW 004, Kecamatan Batu Ampar, Kabupaten Kubu Raya, Provinsi Kalimantan Barat, Indonesia, 78385
Telephone	-
Email	-

1.6 Other Entities Involved in the Project

Organization name	Bentang Kalimantan
Role in the project	Technical Advisor
Contact person	Anton P. Widjaya
Title	Director
Address	Jl. Parit Haji Husein II, Komp. Mutiara Gading No. C3, Kel. Bangka Belitung Darat, Kec. Pontianak Tenggara, Kota Pontianak, West Kalimantan, Indonesia
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Organization name	South Pole
Role in the project	Project Developer
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1.7 Ownership

In February, 2017, the MoEF issued 'Village Forest' management permits to 10 Village Forest management units/lembaga pengelolaan Hutan Desa (LPHDs) in 10 villages in the Padang Tikar Landscape through a social forestry program. The mandate given in the permit is for each LPHD to carry out forest conservation, protection, and management and the utilization of ecosystem services and sustainable non-timber forest products. The Village Forest permits require LPHDs to act as a representative of the village and manage the Village Forest. In the implementation of Village Forest management and, in particular, carbon asset management activities, the 10 LPHDs, along with the village government(s), will be a project proponent and project owner of this project. Bentang Kalimantan will continue to support the implementation of activities in the village.⁵

⁵ The supporting documents are available in the folder Section 1/Ownership.

1.8 Project Start Date

In accordance with the VCS definition, which states that the start date of an AFOLU project is the date on which activities that led to the generation of GHG emission reductions and removals are implemented, the project start date was August 30, 2017.

Ten villages in the Padang Tikar Landscape proposed a social forestry/Village Forest scheme to the MoEF in 2016. The proposals were then approved by the MoEF, with the Village Forest permits for the 10 villages being issued on February 14, 2017. The issuance of permits makes it possible for local communities to develop sustainable forest management in forest areas around the village. A key component of the sustainable forest management activities developed by local communities will be the utilization of environmental services, including the development of carbon management programs. In other words, the issuance of permits is the first milestone for communities working toward reducing emissions from activities that threaten forests, such as illegal logging, wildfires, and the draining of peatlands. The first implemented activity in the Padang Tikar Landscape after receipt of the Village Forest permits was mangrove revegetation activities in Nipah Panjang village. The start date of the project activity was August 30, 2017,⁶ or the day that the district government of Kubu Raya opened a new mangrove tourism destination in the Padang Tikar Landscape.⁷

1.9 Project Crediting Period

The duration of the VCS project crediting period is 30 years, with the project beginning on 30-August-2017 and ending on 29-August-2047.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

When completing a draft project description for the purpose of listing on the pipeline as under development, complete the following information; otherwise, delete this text.

The estimated annual GHG emission reductions/removals of the project are:

- ☐ <20,000 tCO₂e/year
- ☐ 20,000 – 100,000 tCO₂e/year
- ☒ 100,001 – 1,000,000 tCO₂e/year
- ☐ >1,000,000 tCO₂e/year

⁶ The supporting documents are available in the folder Section 1/Project Start Date.

⁷ Available at: <https://www.youtube.com/watch?v=x8Q-W9Ly8Q>.

Project Scale	
Project	
Large project	X

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
2017	259,516
2018	424,023
2019	460,722
2020	493,574
2021	539,971
2022	576,589
2023	597,042
2024	638,400
2025	677,725
2026	728,784
2027	509,226
Total estimated ERs	5,905,572
Total number of crediting years	10
Average annual ERs	590,557

1.11 Description of the Project Activity

The Padang Tikar Landscape project is not located within a jurisdiction covered by a jurisdictional REDD+ program. The project aims to reduce GHG emissions from deforestation by balancing protection, production, and inclusion in Village Forests. This project has the following three outcomes:

- 1) the enhancement of the carrying capacity of critical peat hydrological units in Village Forests;
- 2) the protection of rare, endemic, and endangered wild plants and animals in Village Forests; and

- 3) an increase in community welfare through the LPHD-led development of alternative sustainable income opportunities.

To achieve these outcomes in the Landscape, short-term funding has been secured to support the implementation of activities. The Sustainable Trade Initiative (IDH) was the main donor supporting major activities on the ground in the Landscape. The project activities have been designed to reduce the threat of wildfires, illegal logging, and agricultural expansion in Village Forests by implementing a community livelihood approach. Figure 1 below describes the project's theory of change. It develops a long-term vision for the Padang Tikar Landscape and aims to address underlying causes and direct drivers of deforestation in the Landscape.

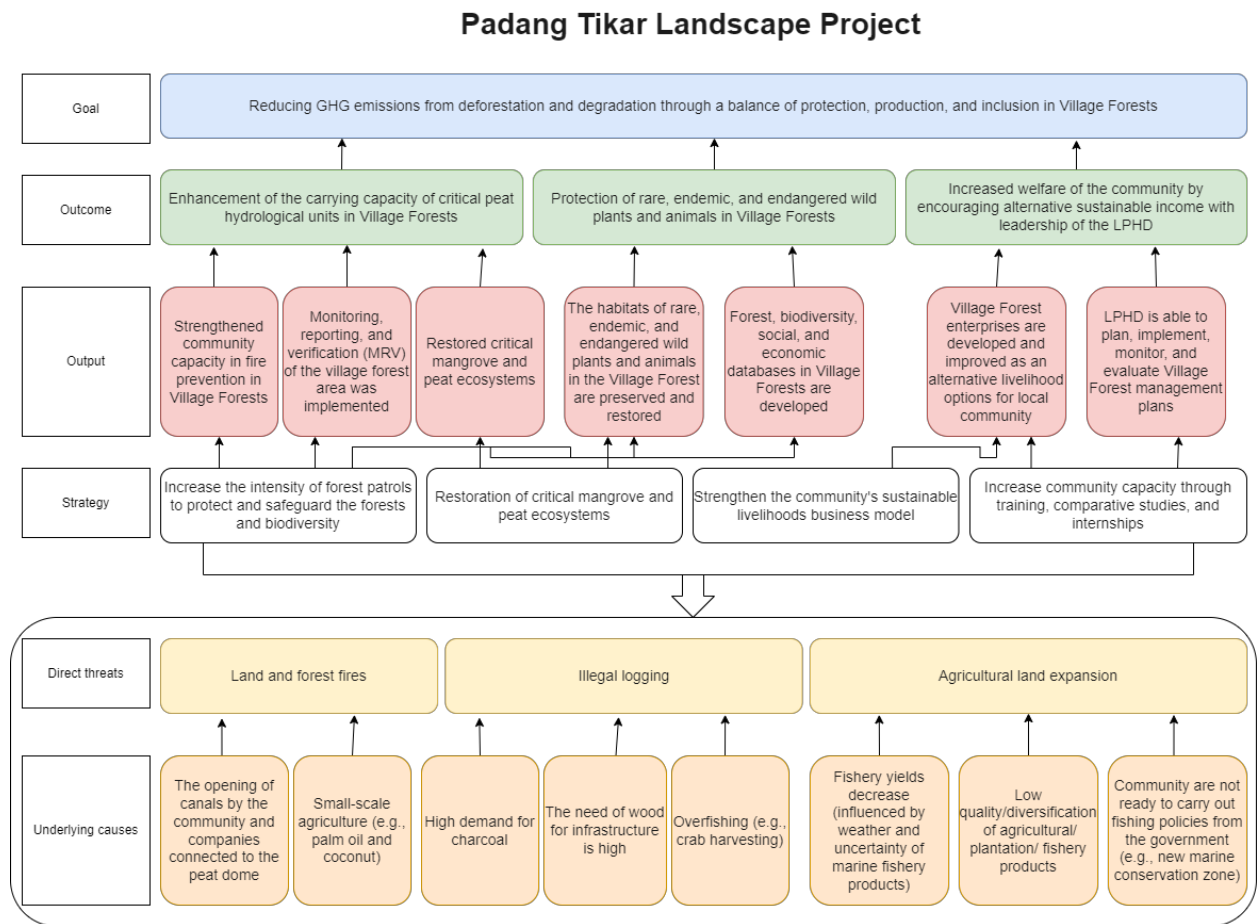


Figure 1: Theory of change in the Padang Tikar Landscape project

The project activities implement four strategies, as described in the following section:

Increasing the Intensity of Forest Patrols to Protect and Safeguard the Forest and its Biodiversity

This strategy covers the project activities that aim to secure the protected areas in the Padang Tikar Village Forest by reducing the risks of land and forest fires, illegal logging, and forest conversion due to small-scale agricultural expansion.

This strategy includes the following activities:

- 1) developing the Forest Guard and the Fire Prevention Unit system and its standard operating procedure to safeguard the protected forest area and other protected areas;
- 2) forest protection training for the Forest Guard and the Fire Prevention Unit; and
- 3) community patrols to protect forests and other areas within the Village Forest.

To support the implementation of this strategy, an analysis of the Landscape's deforestation rate over a period of 16 years (i.e., 2000–2016) has been conducted at a scale of 1:50.000 to provide baseline information for the PA. Furthermore, a WebGIS system (<http://www.pantauhutandesa.com/>) has been developed to support the implementation of monitoring activities for the year 2019-2020 (Figure 2).

WebGIS is a web-based smart patrol system developed to analyze land-use change. It includes an alert system designed to measure the potential impact of an activity on land-use change within the Landscape. If there are potential impacts on land-use change or deforestation, ground-checking will be conducted using GPS and unmanned aerial vehicles or drones to collect evidence. The results of this activity will then be used as the basis for further action. This system ensures that each activity contributing to deforestation and degradation can be detected quickly and periodically monitored.

By involving stakeholders and communities within the Landscape, the WebGIS system can provide accountability and transparency regarding deforestation activities. Moreover, this system is accessible to everyone, including stakeholders outside of the Landscape. To facilitate greater community involvement, a series of training sessions for communities and stakeholders will need to be conducted to build capacity on forest monitoring tools. This training will help build stakeholder inclusiveness in the system.

This strategy contributes to the outcomes of this project by reducing the quantity of GHG emissions released from peat fire activities, illegal logging, and land clearing for small-scale agriculture.

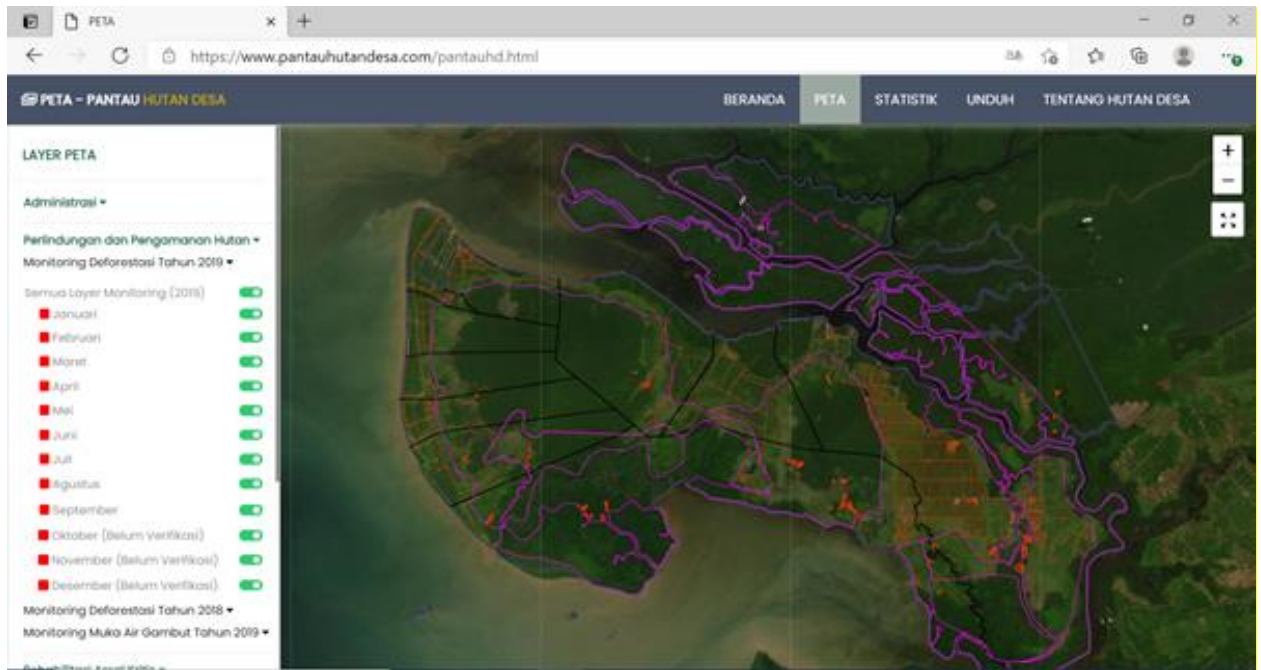


Figure 2: The WebGIS system (was active on the year 2019-2020)

Restoration of Critical Mangrove and Peat Ecosystems

This strategy covers the project activities that aim to restore the ecological function of critical mangrove and peat ecosystems in the Padang Tikar Village Forest. This strategy contributes to the project objective of recovering the hydrological function of peat and mangrove ecological systems, increasing biodiversity in the area in the long term.

Several activities in this strategy include:

- 1) rewetting 150 ha of degraded peat ecosystem using a peat hydrological system approach;
- 2) rehabilitating 500 ha of degraded peat and mangrove areas using a natural succession approach; and
- 3) installing devices to monitor peat water levels. Four villages (Ambarawa, Sungai Besar, Tasik Malaya, and Padang Tikar 1) have already installed devices monitoring peat water levels.

This strategy began with a mapping activity to develop restoration targets and plans for the 10 Village Forests, spanning 2018–2028. Several actions should now be taken, including reaching the necessary agreements with local communities around these targets and plans, installing signposts and peat water monitoring devices in the restoration areas, preparing nursery facilities for agroforestry, and conducting pruning activities in the natural succession area under the rehabilitation program.

The agroforestry activity in the peatland area will be conducted with an emphasis on commodities valuable to each local community, such as Porang, Kaliadra, coffee, Moringa, and others. This activity also includes building the local community's capacity on the agroforestry system.

The mangrove restoration activities will also be carried out in seven villages in the landscape: Padang Tikar 1, Sungai Besar, Tanjung Harapan, Nipah Panjang, Sungai Jawi, Medan Mas, and Batu Ampar. The planting will be conducted on degraded land that has been previously mapped.⁸ This activity involves the communities as primary stakeholders and not only seeks their involvement, but also aims to increase awareness of the importance of improving the ecological function of mangroves in the area.

Strengthening the Community's Sustainable Livelihood Business Model

The threats of deforestation and degradation in the mangrove and peatland ecosystems occur due to limited livelihood options for people in the communities surrounding the Landscape. A reduction in sea-based products due to overfishing has led to many fishermen turning to land-based livelihoods, such as coconut and palm oil production, both of which cause land clearing and contribute to an increase in peat fires. High demand for charcoal from the domestic market and international buyers has caused illegal logging in the protection area and has led communities to become dependent on the exploitation of the Landscape's extensive mangrove forests. The community's limited knowledge of and access to livelihood opportunities beyond these extractive activities has increased deforestation in the Landscape. To address this deforestation, reduce degradation, and increase community welfare, the sustainable livelihood business model has been introduced in the Landscape.

Four main business models have been introduced in the Landscape:

- 1) sustainable crab production;
- 2) sustainable honey production;
- 3) sustainable coconut production; and
- 4) sustainable wood and coconut charcoal production.

Business models for sustainable crab and honey production

These business models assume that the greater income generated through sustainable crab and honey production will attract those currently working in mangrove charcoal production. The initial investment needed for these models has currently been secured through a partnership with the Forest Development Financing Center Public Service Agency and the MoEF (BLU-P3H MoEF) and is strengthened through collaboration with PT KNJ and Credit Union KNB.

⁸ The supporting documents are available in the folder Section 1/Project Activity

The BLU-P3H MoEF is a government body that provides loans to local communities that can be used to strengthen commodity business models in the production stage at village level. Collaboration with companies like PT KNJ and Credit Union KNB will strengthen commodity business models at the marketing stage, including those related to adding value to products, distribution, and markets, and expanding the number of financial institutions that the community can access.

Several activities in this strategy include:

- 1) increasing co-investment/loan deals for community development (business model);
- 2) escorting the business unit in developing production infrastructure based on a co-investment agreement and increasing production targets for honey and crab; and
- 3) improving market assurances to maintain commodity production based on co-investment.

Business model for sustainable wood and coconut charcoal production

This business model assumes that fully transitioning local communities away from livelihoods based on the extraction of mangrove charcoal will take time. Sustainable wood charcoal production is an alternative that can be implemented to reduce the negative effects of charcoal production in mangrove areas. Wood charcoal production, essential to the Landscape's communities since the 1920s, is one of the primary drivers of deforestation in the Village Forest. Approximately 400 units of traditional burning houses have been identified, alongside roughly 500 ha of cutting area inside the protected forest area, part of the Village Forest boundary. Therefore, two strategies need to be achieved: 1) encouraging the legal recognition of mangrove cutting areas under the social forestry program; and 2) improving charcoal production and distribution chains. However, there is currently no support from the MoEF for either strategy.

Several activities in this strategy include:

- 1) engaging with other stakeholders to identify solutions related to sustainable wood charcoal production in mangroves; and
- 2) developing a cooperative/business unit for other potential commodities, such as coconut copra and coconut charcoal, within the Village Forest area. Currently, two cooperatives for honey and coconut have been developed, in addition to 20 social forestry business units/*Kelompok Usaha Perhutanan Sosial* (KUPS) for honey, crab, and coconut production.

Increase Community Capacity Through Training, Comparative Studies, and Internships

This strategy covers project activities that aim to increase the community's awareness and knowledge of the importance of forests, biodiversity, and sustainable livelihoods. Such activities

will contribute to maintaining the ecological function of peat and mangrove ecosystems, while ensuring that locals are still able to derive value from the forests.

Several activities in this strategy include:

- 1) developing community-based mangrove learning and a monitoring center at Mangrove Mega Biodiversity;
- 2) training instructors in commodity production for honey and crab;
- 3) training business units in managerial and financial skills to manage the co-investment deal with BLU-P3H MoEF for honey and crab production; and
- 4) training on effective and efficient business management and development.

The assumption under this strategy is that, as the local community's knowledge and awareness of the importance of forests, biodiversity, and sustainable livelihood increases, the threat of future deforestation and degradation will be reduced, and local livelihoods will become more sustainable.

1.12 Project Location

The Padang Tikar PA is located on the western side of the island of Kalimantan. Figure 3 shows the location of the PA alongside the location of the villages surrounding the PA. The Padang Tikar Landscape consists of 11 villages, although only 10 villages, namely Batu Ampar, Teluk Nibung, Nipha Panjang, Medan Mas, Padang Tikar 1, Tasik Malaya, Sungai Besar, Sungai Jawi, Ambarawa, and Tanjung Harapan, are involved in the project as they have been granted the Village Forest permit from the MoEF. The expansion landscape of the project is at coordinates 0°37'2.11" to 0° 0'13.18" South latitude and 109°15'26.49" to 109°44'53.91" East longitude, and covers about 58,672.7 ha, of which 1,210.3 ha, 32,337.4 ha, and 25,125.1 ha corresponded to secondary dryland forest, secondary mangrove forests, and secondary swamp forests respectively in 2017 (i.e., the project start date).

Table 1: Village Forest coordinate points

Village	East longitude	South latitude
Ambarawa	109°18'49.8"	0°49'52.6"
Batu Ampar	109°18'49.8"	0°46'59.8"
Medan Mas	109°18'31.1"	0°43'3.4"
Nipah Panjang	109°23'59.3"	0°45'3.4"
Padang Tikar I	109°15'24.3"	0°41'45.4"
Sungai Besar	109°15'26.7"	0°45'41.8"

Village	East longitude	South latitude
Sungai Jawi	109°18'20.9"	0°48'4.9"
Tanjung Harapan	109°27'30.0"	0°52'36.9"
Tasik Malaya	109°17'10.8"	0°44'49.9"
Teluk Nibung	109°29'49.7"	0°48'2.7"

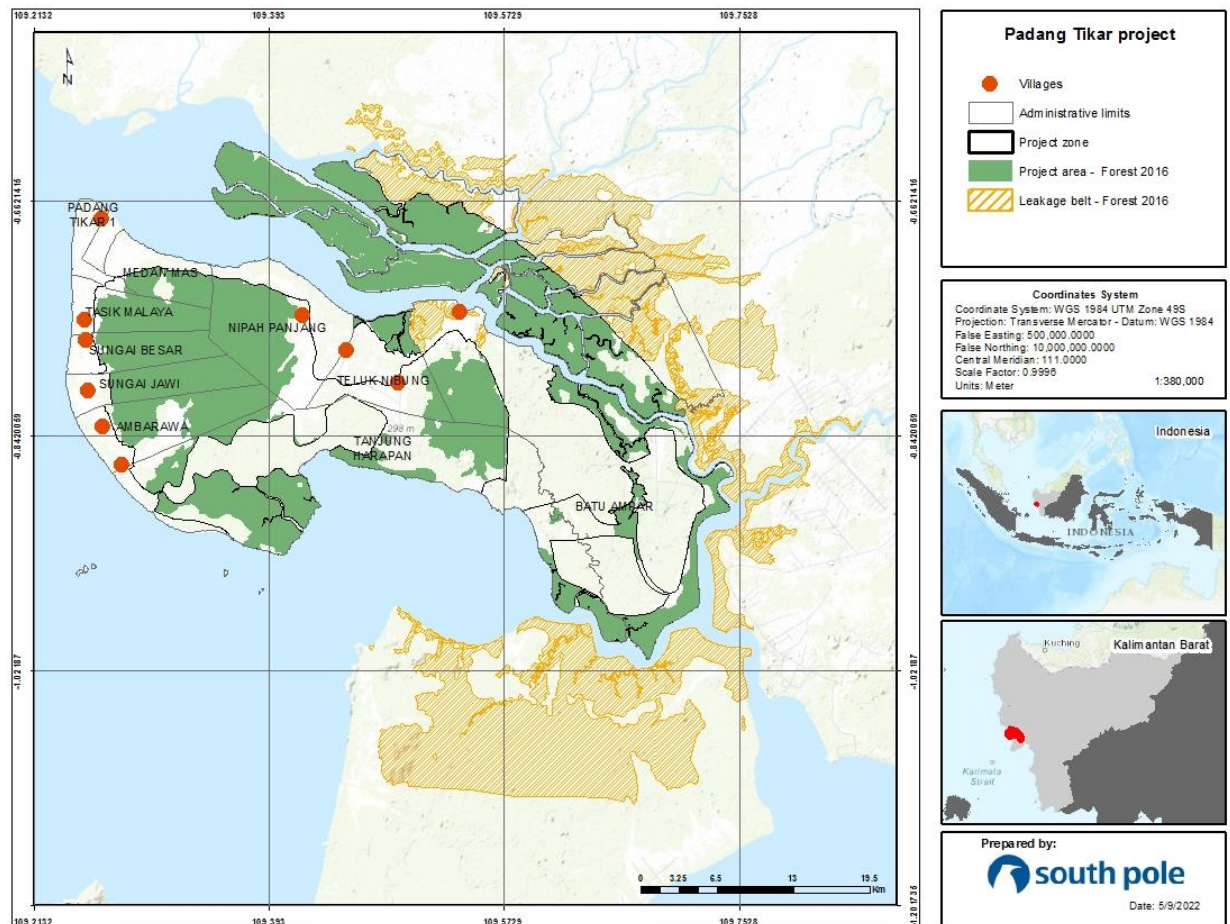


Figure 3: Padang Tikar Landscape location and boundary

1.13 Conditions Prior to Project Initiation

- Ecosystem type:** the ecosystem types in the Padang Tikar Landscape project are classified into three types, namely secondary dryland forests, secondary swamp forests, and secondary mangrove forests.

- **Current and historical land-use:** the baseline scenario is the same as the conditions which existed prior to the project initiation Please refer to section 3.4 for further detail on the baseline scenario.
- **Has the land been cleared of native ecosystems within 10 years of the project start date?**

☐ Yes

☒ No

This section aims to provide an overview of the physical parameters of the Padang Tikar Landscape PA. It is important to acknowledge the social factors which also contribute to shaping the physical landscape. Indigenous communities have long resided in these areas, and have thus influenced the Landscape whilst playing a key role in its stewardship.

Topography

The Padang Tikar Landscape PA has, historically, been densely forested, consisting of various types of forest ecosystem, including swamp forests (e.g., coastal mangrove forests and peatlands) and inland forests (e.g., inland peatland and dryland forests). Its inland forests are composed of mineral soil. Well-drained lowland dipterocarp forests are intermixed with swamp forests. In general, Kalimantan's remaining intact forests are found mainly at higher elevations, while its forests in the lowlands and wetlands, including agroforestry plantations, scrublands, and croplands, are fragmented forests.

Hydrology

The mangrove biosphere in the archipelago lies across three subdistricts within the Kubu Raya district, these being Batu Ampar, Kubu, and Teluk Pakedai.⁹ Batu Ampar sits north of the Padang Tikar area, with 21 true mangrove species and 17 associate mangrove species found in this region.¹⁰ These include endemic species such as *Kandelia candel*, *Rhizophora spp.*, *Bruguiera spp.*, and *Nypa fruticans*.¹¹ Within the Padang Tikar area, there has at times been rapid growth in the *Rhizophora macronata* and *R. apiculata* mangrove species.¹² Some parts of

⁹ Mustikasari, R., Prasetiamartati, B., Santoso, N., Syah, C., & Tai, H.S. (2008), "Mangrove Forest and Charcoal Production: Case of Batu Ampar, West Kalimantan." In a paper submitted for the IASC 2008 Global Conference.

¹⁰ Mustikasari, R., Prasetiamartati, B., Santoso, N., Syah, C., & Tai, H.S. (2008), "Mangrove Forest and Charcoal Production: Case of Batu Ampar, West Kalimantan." In a paper submitted for the IASC 2008 Global Conference.

¹¹ Ibid.

¹² Asian Wetland Bureau, (1992), "Environmental Impact Assessment Integrated Swamps Development Project (Riau, Jambi and West Kalimantan)", Indonesia: Integrated Swamps Development Project, World Bank, p.38, <https://documents1.worldbank.org/curated/pt/516591468772494515/pdf/multi-page.pdf>.

the coast are covered by mangrove swamps, while the rest is covered by nipah (*Nypa fruticans*).¹³

West Kalimantan has been described as the “thousand rivers region,”¹⁴ due to the large number of small but accessible rivers that flow through the region. In 1977, a hydrographic survey conducted in the area discovered that the tides around Padang Tikar are diurnal, with a range of up to 2 meters (m) or more in a West/Northwest direction.¹⁵ Padang Tikar itself sits on an estuary whereby ocean embayments are diluted by runoff from the land.¹⁶ Within these river systems, aquatic fauna, such as fish, crustaceans, gastropods, bivalves, polychaetes, phytoplankton, and zooplankton, are the dominant organisms.¹⁷

According to the World Bank (1992), the tidal swamps within the Padang Tikar area are of the 'Kajapah' type.¹⁸ Kajapah systems are characterized by low and vast coastal plains with a slope under 2% and a relief difference below 2 m.¹⁹ They are found under halophytic vegetation with sedimentary soil,²⁰ and contain rock and mineral systems dominated by alluvium with a mix of young/recent estuary and marine materials.²¹

¹³ Pardjaman, D. (1977). “SOME NOTES OF THE HYDROGRAPHIC SURVEYS IN INDONESIAN WATERS”, Marine Research in Indonesia, 20, p. 142.

¹⁴ Akbar, M.F. (2008). “Profil dan Permasalahan Sistem Logistik Sungai di Kabupaten Kubu Raya Provinsi Kalimantan Barat”. *Jurnal TIN Universitas Tanjungpura*, 1(1).

¹⁵ Pardjaman, D. (1977). “SOME NOTES OF THE HYDROGRAPHIC SURVEYS IN INDONESIAN WATERS”, Marine Research in Indonesia, 20, p. 143.

¹⁶ Ibid.

¹⁷ Mustikasari, R., Prasetyamartati, B., Santoso, N., Syah, C., & Tai, H.S. (2008), “Mangrove Forest and Charcoal Production: Case of Batu Ampar, West Kalimantan.” In a paper submitted for the IASC 2008 Global Conference.

¹⁸ Asian Wetland Bureau, (1992), “Environmental Impact Assessment Integrated Swamps Development Project (Riau, Jambi and West Kalimantan)”, Indonesia: Integrated Swamps Development Project, World Bank, p.28, <https://documents1.worldbank.org/curated/pt/516591468772494515/pdf/multi-page.pdf>.

¹⁹ Diana, R., Seftianingrum, R., Suhardiman, A., Sulistioadi, Y. B., & Suwasono, R. A. (2020). Floral Composition of the Kayan-Sembakung Delta in North Kalimantan (Indonesia) in Different Disturbance Regimes. *Journal of Coastal Research*, 36(4), 742. <https://www.jstor.org/stable/26928927>.

²⁰ Darmawan, M., Hashim, M., Helmi, M., Qin, R., Rahadiati, A., Shih, P.T.Y., Sutrisno, D., Yusmur, A., & Zhang, L.. (2021). Spatial-Planning-Based Ecosystem Adaptation (SPBEA): A Concept and Modeling of Prone Shoreline Retreat Areas. *ISPRS International Journal of Geo-Information*, 10(3), p.3.

²¹ Ibid.

Table 2, taken from the International Society for Photogrammetry and Remote Sensing's (ISPRS) *International Journal of Geo-Information*²² elaborates on the typical characteristics of Kajapah land types:

Table 2: Characteristic of Kajapah

Parameters	Kajapah characteristic	
Physiographic	Slope (m)	<2
	Elevation (%)	<2
Morphology		Flat
Landform		Swamp
Genesis		Marine
Land use		Aquaculture, mangroves
Geology	Lithology	Sedimentary
	Rock type	Alluvium
		Recent estuarine marine
Climate	Rainfall (mm)	400–3100
	Wet months	2–10
	Dry months	0–7
Soil	Solum (cm)	>150
	Texture	Sandy loam to clay loam
	pH	7
Drainage		Poor drainage
Groundwater depth (cm)		0–45
Salinity		Moderately saline
		Brackish fisheries
Suitability		Ponds
		Mangroves

²² Darmawan, M., Hashim, M., Helmi, M., Qin, R., Rahadiati, A., Shih, P.T.Y., Sutrisno, D., Yusmur, A., & Zhang, L.. (2021). Spatial-Planning-Based Ecosystem Adaptation (SPBEA): A Concept and Modeling of Prone Shoreline Retreat Areas. *ISPRS International Journal of Geo-Information*, 10(3), p.10.

Soils

The project site lies within the West Kalimantan region, an area rich in peatland landscapes essential for water regulation, carbon storage, and biodiversity.²³ Human activities in the West Kalimantan region have threatened peat swamp forest, largely due to growth in agricultural land use and forest production.²⁴ Peat in Kalimantan is thinner and contains a carbon average of about 2000 megagrams (Mg)/ha. Additionally, the carbon content of mineral soil in the area is concentrated in the few centimeters (cm) closest to the surface layer, and rarely exceeds 250 Mg/ha.²⁵

The soils identified by geographic information system (GIS) analysis were Haplic Sulfaquents, Sulfic Fluvaquents, Typic Haplohemists, Typic Kandiodults, Typic Quartzipsamments, and Typic Sulfaquents. They fall into the categories of entisols, inceptisols, histosols, and ultisols (see Figure 4).

Entisols are commonly found in areas of high erosion, quartz sand, and wetland. They can also be created by disturbing the land, as occurs with extraction, the moving of earth materials, or the disposal of waste products. They differ from inceptisols by a lesser subsurface accumulation of transported clay.²⁶ Inceptisols are commonly found with underlying weathering-resistant parent materials (e.g., quartzite or siliceous sandstone) or in topographic settings conducive to erosion or waterlogging.²⁷ Histosols are typically formed under waterlogged conditions and contain a high organic content. After drainage for agricultural use, the organic material is prone to oxidation, leading to fire hazards and subsidence.²⁸ Ultisols are

²³ Agus, Fahmuddin, et al. (2010). "Carbon budget and management strategies for conserving carbon in peatland: case study in Kubu Raya and Pontianak districts, West Kalimantan, Indonesia." *Proceeding of International Workshop on Evaluation and Sustainable Management of Soil Carbon Sequestration in Asian Countries*. Bogor, Indonesia: Indonesian Soil Research Institute. p.217.

²⁴ Aspirin, A., & Nusantara, R.W. (2017). "Differentiation of soil organisms at different types of peatland in West Kalimantan, Indonesia." *International Journal of Bonorowo Wetlands*, 7(1), p.26. <https://www.smujo.id/bw/article/view/1774/1695>.

²⁵ Agus, Fahmuddin, et al. (2010). "Carbon budget and management strategies for conserving carbon in peatland: case study in Kubu Raya and Pontianak districts, West Kalimantan, Indonesia." *Proceeding of International Workshop on Evaluation and Sustainable Management of Soil Carbon Sequestration in Asian Countries*. Bogor, Indonesia: Indonesian Soil Research Institute. p.218.

²⁶ Britannica, The Editors of Encyclopaedia. "Entisol". *Encyclopedia Britannica*, December 16, 2011, <https://www.britannica.com/science/Entisol>

²⁷ Britannica, T. Editors of Encyclopaedia. "Inceptisol." *Encyclopedia Britannica*, March 2, 2012. <https://www.britannica.com/science/Inceptisol>

²⁸ Britannica, T. Editors of Encyclopaedia. "Histosol." *Encyclopedia Britannica*, June 29, 2011. <https://www.britannica.com/science/Histosol-soil>

characterized by a humus-rich surface horizon,²⁹ a layer of clay below, and a nutrient content low in available calcium, magnesium, potassium, and sodium.³⁰

Information on the specific soils identified is as follows:

- 1) Haplic Sulfaquents have, in one or more horizons at a depth 20–50 cm below the mineral soil surface, either or both less than eight percent clay in the fine-earth fraction or an n value³¹ of 0.7 or less;³²
- 2) Sulfic Fluvaquents have sulfidic materials at a depth 50–100 cm below the mineral soil surface;³³
- 3) Typic Haplohemists do not have a lithic contact and do not have a layer of water within the control section below the surface tier. They are used mainly as woodland, cropland, or wildlife habitat;³⁴
- 4) Typic Kandiodults are freely drained soils that do not have a sombric horizon and are very deep. They possess a sandy clay loam texture. These soils have less than five percent plinthite and a low cation-exchange capacity. They do not have a surface mantle or layer in the upper 75 cm that has both a low bulk density and a high content of weakly crystalline minerals;³⁵
- 5) Typic Quartzipsamments are soils that have a udic or perudic moisture regime, have deep ground water in all seasons, do not have distinct color horizons, and have at least moderately active clays or very little clay. Most are used as forest, but some are used as cropland or pasture;³⁶ and

²⁹ Approximately 60 percent carbon, 6 percent nitrogen, and smaller amounts of phosphorus and sulfur

³⁰ Britannica, T. Editors of Encyclopaedia. "Ultisol." Encyclopedia Britannica, September 27, 2012.
<https://www.britannica.com/science/Ultisol>

³¹ The n value characterizes the relation between the percentage of water in a soil under field conditions and its percentages of inorganic clay and humus. The n value is helpful in predicting whether a soil can be grazed by livestock, whether it can support other loads, and what degree of subsidence would occur after drainage.

³² Soil Survey Staff. (1999). 'Soil Taxonomy.' US Department of Agriculture Handbook No. 436. 2nd edn. US Government Printing Office: Washington, D.C., p.403. https://www.nrcs.usda.gov/Internet/FSE_DOCUMENTS/nrcs142p2_051232.pdf

³³ Ibid, p.400.

³⁴ Ibid, p.482.

³⁵ Soil Survey Staff. (1999). 'Soil Taxonomy.' US Department of Agriculture Handbook No. 436. 2nd edn. US Government Printing Office: Washington, D.C., p.756.

³⁶ Ibid, p.435.

- 6) Typic Sulfaquents are soils that have, in all horizons at a depth 20–50 cm below the mineral soil surface, both an n value of more than 0.7 and eight percent or more clay in the fine-earth fraction. They are nearly level and support mostly grassy and forb vegetation. Most of the soils are used as wildlife habitats.³⁷

In addition, a study by Grealish and Fitzpatrick (2013) found high organic carbon concentrations in Haplic Sulfaquents in the surface layers and in deeper subsoil layers in Sulfic Fluvaquents.³⁸

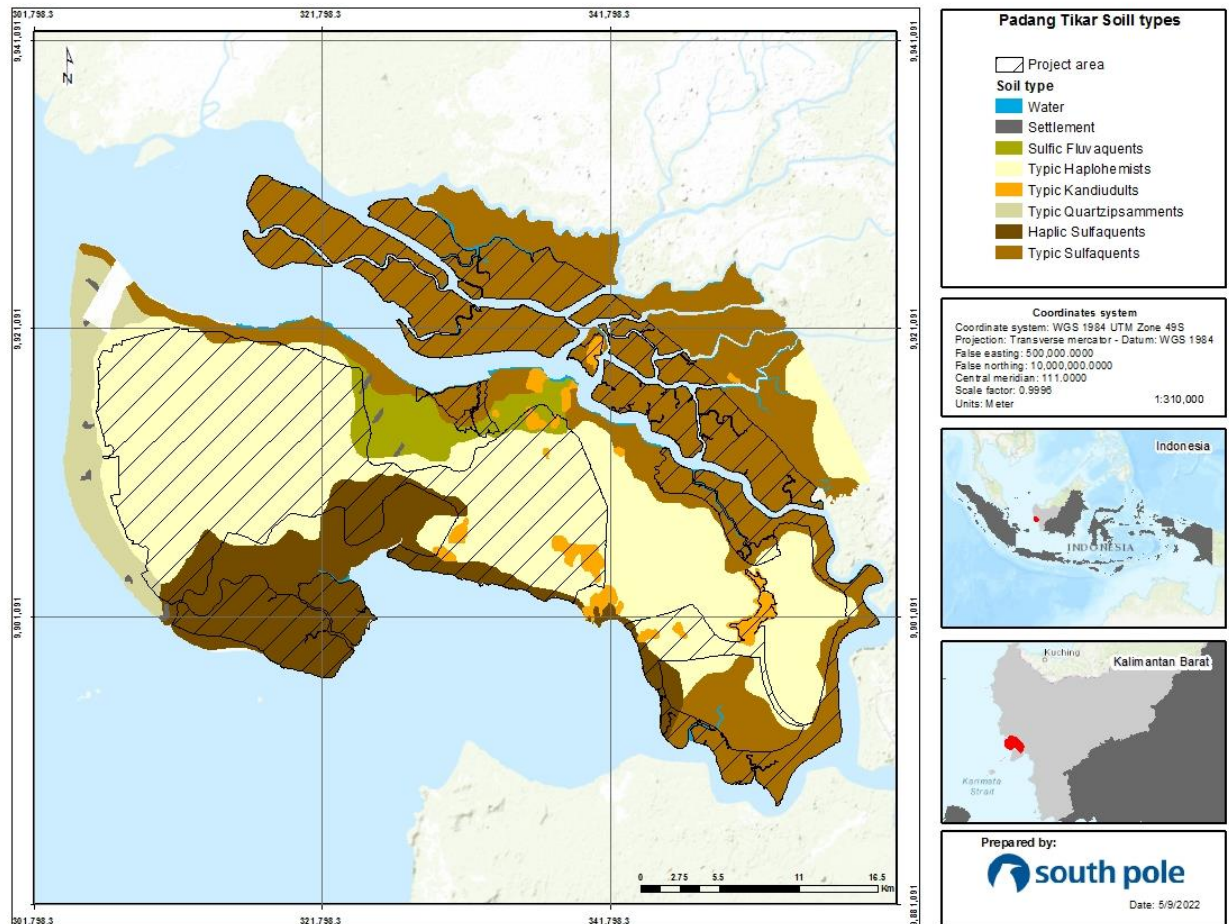


Figure 4: Soil types in the Padang Tikar Landscape

³⁷ Ibid, p.403.

³⁸ Fitzpatrick, .R.W., & Grealish, G.J. (2013). Acid sulphate soil characterization in Negara Brunei Darussalam: a case study to inform management decisions. p.435.

https://bsssjournals.onlinelibrary.wiley.com/doi/epdf/10.1111/sum.12051?saml_referrer

Climate

The island of Kalimantan, where the Padang Tikar Landscape is located, has a tropical climate, meaning that temperatures average roughly 28°C (although this can be as low as 15°C in the inland highlands), with high humidity for most of the year. Like other countries in the region, Indonesia's seasons vary from the dry season, often occurring between July and September, and the rainy season, often between October and May.³⁹ In the rainy season, there can be heavy rainfall (i.e., monsoons) and, on occasion, typhoons, although the latter is rarer in Kalimantan. Kalimantan, and especially its northwestern regions, experiences relatively high rainfall compared to other parts of the archipelago and the rest of the world, with the interior of Kalimantan estimated to have more than 2,000 millimeters (mm) of rainfall per year.⁴⁰ Climate change has played a significant role in its forest conditions, such as causing a prolonged dry season and heavy rain during the rainy season. Extended dry periods pose the risk of forest fires, which have seen notable occurrences in 1997, 2006, and 2015. The 1997 forest fires, in particular, occurred and were exacerbated as a result of the El Niño-Southern Oscillation. It is anticipated that the effects of El Niño will become more extreme as a result of climate change.

Vegetation

Crops cultivated by settlements in the Padang Tikar Landscape suggest that the land is arable. Most agricultural land-use in the Padang Tikar Landscape is for growing rice, with almost 88% of agricultural land in West Kalimantan (e.g., peatlands and wetlands, though this has also extended to include cleared forested land) being used for rice. Other crops that are cultivated include coconut, copra, rubber, bananas, and rambutan. There is also a history of growing cassava, corn, and coffee in West Kalimantan, so it is possible that these are also grown within the Padang Tikar Landscape. Oil palm also grows well in Kalimantan, as forested areas have also been cleared for oil palm plantations.

The Padang Tikar Landscape consists of three types of forests, including secondary dryland forest, secondary swamp forest, and secondary mangrove forest, as shown in Table 3 and Figure 5.

Table 3: Forest types and areas in project area

Forest type	Area (ha)	Coverage (%)
Secondary dryland forest	1,210.3	2.1
Secondary swamp forest	32,337.4	55.1
Secondary mangrove forest	25,125.1	42.8
Total	58,672.7	100.0

³⁹ Available at: https://factsanddetails.com/indonesia/Nature_Science_Animals/sub6_8a/entry-4079.html.

⁴⁰ Available at: https://factsanddetails.com/indonesia/Nature_Science_Animals/sub6_8a/entry-4079.html.

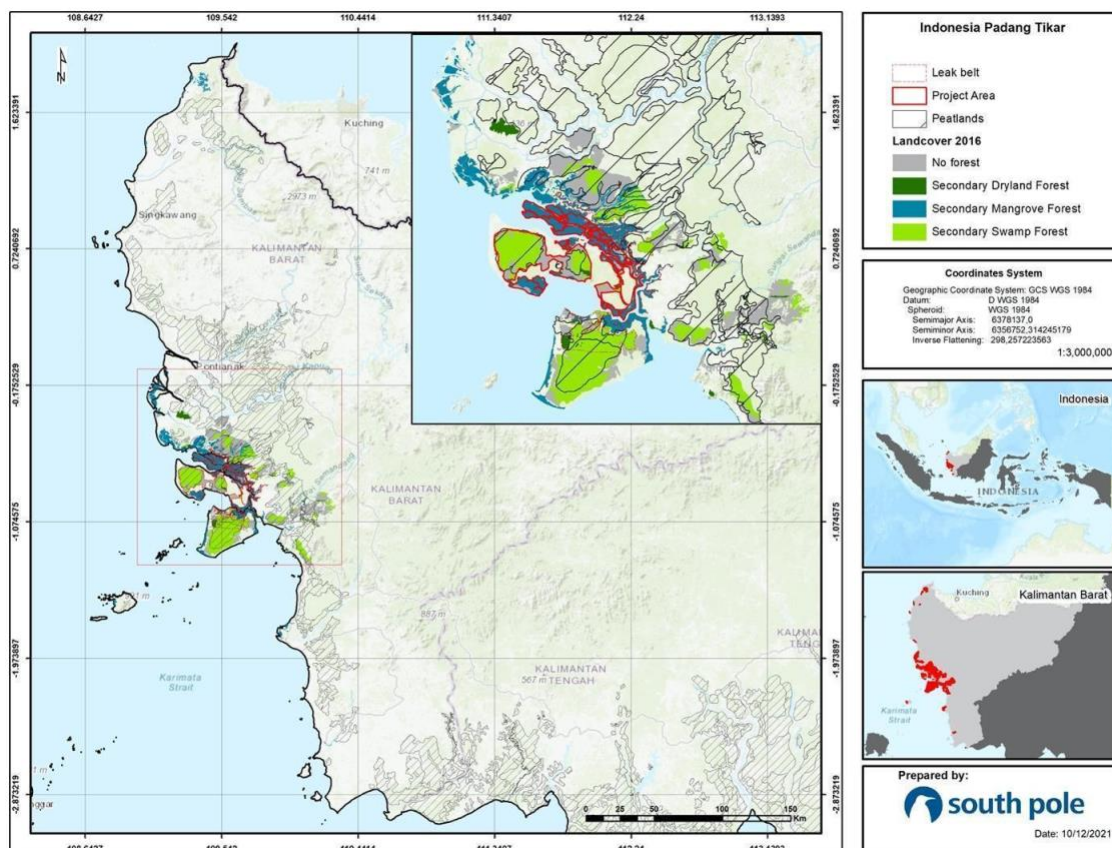


Figure 5: Forest types in the Padang Tikar Landscape

The project has a very rich and complete diversity of mangrove ecosystems. The mangrove formation in this project consists of *Avicennia*, *Ceriops*, *Sonneratia*, *Rhizophora*, *Bruguiera*, *Xylocarpus*, and *Nypa*. According to the study by SAMPAN Kalimantan in 2016, Padang Tikar Landscape contains 33 mangrove species, as shown in Table 4.

Table 4: Dominant mangrove species in Padang Tikar Landscape

Scientific name ⁴¹	The IUCN Red List of Threatened Species	Global status
<i>Acanthus ebracteatus</i>	LC	Decreasing
<i>Acanthus ilicifolius</i>	LC	N/A
<i>Acrostichum aureum</i>	LC	Stable
<i>Acrostichum speciosum</i>	LC	Stable

⁴¹ The supporting documents are available in the folder Section 1/Forest Tree Type/Sustainable Mangrove Management for Community.

Scientific name ⁴¹	The IUCN Red List of Threatened Species	Global status
<i>Aegiceras corniculatum</i>	LC	Decreasing
<i>Amyema anisomeres</i>	N/A	N/A
<i>Avicennia alba</i>	LC	Decreasing
<i>Avicennia marina</i>	LC	Decreasing
<i>Avicennia officinalis</i>	LC	Decreasing
<i>Bruguiera cylindrica</i>	LC	Decreasing
<i>Bruguiera gymnorhiza</i>	LC	Decreasing
<i>Bruguiera sexangular</i>	LC	Decreasing
<i>Bruguiera parviflora</i>	LC	Decreasing
<i>Ceriops decandra</i>	NT	Decreasing
<i>Ceriops tagal</i>	LC	Decreasing
<i>Excoecaria agallocha</i>	LC	Decreasing
<i>Heritiera globosa</i>	EN	Decreasing
<i>Heritiera littoralis</i>	LC	Decreasing
<i>Kandelia candel</i>	LC	Decreasing
<i>Lumnitzera littorea</i>	LC	Decreasing
<i>Lumnitzera racemosa</i>	LC	Decreasing
<i>Nypa fruticans</i>	LC	N/A
<i>Rhizophora apiculata</i>	LC	Decreasing
<i>Rhizophora mucronata</i>	LC	Decreasing
<i>Rhizophora stylosa</i>	LC	Decreasing
<i>Sonneratia alba</i>	LC	Decreasing
<i>Sonneratia caseolaris</i>	LC	Decreasing
<i>Sonneratia ovata</i>	NT	Decreasing
<i>Sarcolobus glubosa</i>	N/A	N/A
<i>Xylocarpus granatum</i>	LC	Decreasing
<i>Xylocarpus moluccensis</i>	LC	Decreasing
<i>Bruguiera hainesii</i>	CR	Decreasing

Note: CR: Critically Endangered, EN: Endangered, LC: Least Concern, NT: Near Threatened, N/A: no data

Biodiversity

Kalimantan has, in general, the highest deforestation rate in the world. In the 17-year period between 2000–2017, there was 6.04 million ha of deforestation, mostly as a result of land expansion driven by the needs of land-based industries. Forests on the west coast of Kalimantan, where the project site is located, and natural forest cover are particularly vulnerable to species extinction. Forested land cover in 2019 saw a significant reduction compared to 2016, decreasing from 57,728 ha to 54,754 ha respectively. These changes occur due to forest and land fires and destructive activities for building needs.

The Padang Tikar Landscape project aims to reverse this trend by protecting sites of internationally recognized importance and species of high conservation value (HCV) in areas of significant peatland, mangrove, and mineral forest ecosystems. This project aims to increase the diversity, integrity, and extent of peat swamp forest, mangrove, and mineral forest ecosystems to increase the abundance, diversity, and critical habitat characteristics of endangered and endemic wildlife.

Table 5: High conservation values identified in the project area

Group	Common name	Species	IUCN Red List of Threatened Species
Fauna	Straw-headed bulbul	<i>Pycnonotus zeylanicus</i>	CR
	Christmas frigatebird	<i>Fregata andrewsi</i>	CR
	Helmeted hornbill	<i>Rhinoplax vigil</i>	CR
	Large-tooth sawfish	<i>Pristis pristis</i>	CR
	Javan pangolin	<i>Manis javanica</i>	CR
	Bornean orangutan	<i>Pongo pygmaeus</i>	CR
	Siamese crocodile	<i>Crocodylus siamensis</i>	CR
	Proboscis monkey	<i>Nasalis larvatus</i>	EN
	Irrawaddy dolphin	<i>Orcaella brevirostris</i>	EN
	Mangrove horseshoe crab	<i>Carcinoscornius rotundicauda</i>	DD
Flora	Berus Mata Buaya	<i>Bruguiera hainesii</i>	CR
	Gedabu	<i>Sonneratia ovata</i>	NT
	Terumtum	<i>Aegiceras floridum</i>	NT

Note: CR: Critically Endangered, EN: Endangered, DD: Data Deficient, NT: Near Threatened

Socioeconomic Conditions

According to Statistics Indonesia, the total population in the PA is 36,844 people (18,768 male and 18,076 female), with an annual population growth rate of 1.03%. Populations are divided into four different age groupings, these being children, teenagers, adults, and elderly, based on data from Statistics Indonesia. The population is dominated by Bugis and Melayu ethnicities, along with migrants from Java Island.⁴²

Table 6: Classification of population in Padang Tikar Landscape

Class	Age	Percentage (%)
Children	0–14	30.13
Teenagers	15–24	16.31
Adults	25–64	49.21
Elders	>65	4.34

The education level of the population in the Padang Tikar Landscape is as follows: 35.30% are elementary school graduates; 15.34% are secondary school graduates; 0.9% are higher education graduates; 31.72% have not attended formal education institutions; and 16.74% are unable to read. Due to the limited availability of educational facilities in the area, people who are eager to continue to higher education (e.g., vocational schools, colleges, or universities) are required to migrate to Pontianak or other cities. Regarding healthcare coverage, 61% of the population in the Landscape are already covered by National Health Care Coverage, while 14% are covered by private health care coverage and 25% are not covered by any health care coverage.

The Padang Tikar Landscape's coastal area is surrounded by an estuary ecosystem, most of which is only accessible via water transportation. Therefore, communities have a high dependency on livelihood activities that are related to natural resource utilization, such as agriculture, plantation, fisheries, and forestry. These sorts of livelihood activities have been implemented among communities for years. The majority of the population's livelihoods are highly related to natural resource utilization, such as the fisheries sector (i.e., fishermen), the agricultural sector (i.e., coconut farmers), the forestry sector (i.e., timber loggers for construction) or woodchar production.

The activities carried out by the community have a correlation to the increase in the deforestation rate. In the Padang Tikar Landscape, factors that cause deforestation include the tendency to change forest cover into another land use, such as for agriculture, commercial plantation, or infrastructure development. Each village has a different main threat towards deforestation. For example, in Batu Ampar, the main threat to forest areas is illegal timber

⁴² Batu Ampar District in Statistics 2019.

harvesting for wood char production, while the main threat to these areas in Teluk Nibung are forest fires as a function of land clearing for agriculture purposes. The agricultural livelihoods mainly adopted by the community are largely related to the establishment of paddy fields. Paddy fields are initially created to fulfill daily needs, with the community only beginning to trade the produce of these fields at the local market once the production rate exceeds these needs. The cultivation system used, traditional cultivation, is highly dependent on rainfall, although irrigation systems have already been implemented in some villages in limited areas.

Another agricultural activity that serves as a significant source of community livelihood is coconut plantation. On average, a single household owns an average of 2 ha of coconut plantation area. Commonly, the coconut is then processed into copra (fruit flesh) or coconut char (fruit shell), with trading prices fluctuating from IDR 7,000–12,000/kilogram (kg) for copra and IDR 2,000–4,000/kg for coconut char. In the fishery sector, people commonly work as traditional fishermen, with 235 effective sailing days annually. The fishing area extends from the surrounding coastal area up to the Karimata strait, with an approximate distance of 125 kilometers (km). Besides work as fishermen, some of the community also implement silvofishery, with the main commodities sourced from this being shrimp and crab.⁴³

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

Relevant laws and regulations on land use, forestry, REDD+, and climate include:

Law No. 6/1994, Ratification of the United Nations Framework Convention on Climate Change

The Indonesian government has ratified the Climate Change Convention through Law No. 6 of 1994, concerning the Ratification of the United Nations Framework Convention on Climate Change (UNFCCC), and is included in Non-Annex I countries. In addition, Indonesia also ratified the Kyoto Protocol to the UNFCCC through Law No. 17 of 2004. Thus, Indonesia is legally bound by obligations and has the right to take advantage of the various opportunities for support offered by the UNFCCC or the UN Framework to achieve the objectives of the Convention. In line with this policy, the Padang Tikar Landscape project is expected to provide support to Indonesia in its efforts to combat climate change.

Regulation No. P.89/Menhut-II/2014 on Village Forests

The MoEF Regulation No. P. 49/2008, which has now been replaced by No. P. 89/2014 on Village Forests, defines “village” as a legal community unit that has territorial boundaries and the authority to regulate and manage the interests of local communities based on origins and customs. These local customs are recognized and respected by the government of Indonesia. The legal terms of reference for a 'Village Forest' are defined by Government Regulation No. 6/2007, with the procedures and requirements for obtaining a Village Forest license as stipulated in Ministerial Regulation Number. P. 49/2008. Through the Village Forest program,

⁴³ Management Strategy of Village Forest to Reduce Deforestation Rate in Batu Ampar District 2019.

village institutions are responsible for protecting forests (Ministerial Regulation No. P. 49/2008, article 34) and reducing the rate of deforestation and degradation. At the same time, Village Forests are expected to be able to guarantee the livelihoods of nearby communities.

Regulation of MoEF No.30 of 2009, Emission Reduction Procedure from Deforestation and Forest Degradation (REDD)

The MoEF Regulation No. P.30/2009, concerning the Mechanisms for Reducing Emissions from Deforestation and Degradation, Chapter III, Article 3, mentions that REDD can be implemented in Village Forests (point i). In addition, Article 4, paragraph 2 also designates the Village Forest holder (manager) as one of the REDD actors at the national level.

Government Regulation No. 71 of 2014, Concerning Protection and Management of Ecosystem of Peat Lands

The protection and management of the peat ecosystem includes planning, utilization, control, maintenance, supervision, and administrative sanctions. The peat ecosystem serves two purposes in protecting the peat environment and cultivating the peat ecosystem. The Indonesian Minister is required to determine the peat ecosystem's protection function for at least 30% of the entire area of the Peat Hydrological Unit, which is located on top of the peat dome and its environs. The plan for protecting and managing the peat ecosystem includes peat ecosystem utilization and/or preservation, alongside function, control, monitoring, maintenance, and protection of the quality of the peat ecosystem, and its adaptation to and mitigation of climate change.

Regulation of the MoEF No. P.83/MENLHK/SETJEN/KUM.1/10/2016, Concerning Social Forestry

This regulation intended to provide guidelines for granting management rights, permits, partnerships, and customary forests in the field of social forestry. Furthermore, the regulation seeks to resolve tenure and justice issues for local people and customary law groups living near forest regions in the interests of community welfare and forest function preservation. Social forestry management uses the principles of justice, sustainability, legal certainty, participation, and accountability. In this project, it is carried out in Village Forests where Village Forest management rights (HPHD) are granted by the Minister to production forests or protected forests that have not been granted a permit, protected forests managed by Perum Perhutani, and certain areas within the forest management unit/*kesatuan pengelolaan hutan* (KPH).

Perdirjen PSKL No. P.11/PSKL/SET/PSL.0/11/2016, Concerning the Application of HPHD Verification Guidelines

This regulation covers the scope of HPHDs, including the procedure for submitting a HPHD application, along with administrative and technical verification procedures. The request for HPHD applications can be implemented in the Indicative Map of Social Forestry Areas (PIAPS), both within the village administration area and Landscape and outside the PIAPS. Technical

verification of HPHD applicants, as referred to in Article 12, is carried out through discussions and interviews with the Chair and members of the LPHD, village heads, community leaders, traditional leaders, and/or the local Camat. These discussions are conducted to discover and share application documents, the applicant's institution, the level of dependence of community livelihoods on the forest, the commodities or types of utilization that the applicant has worked on, and the potential of the area requested.

Perdirjen PSKL No. P.16/PSKL/SET/PSL.0/12/2016, Concerning 'Guidelines for the Preparation of Village Forest Management Plans, Business Work Plans, Community Forest Utilization Business Permits and Business Work Plans for Plantation Forest Timber Products Utilization People'

This regulation aims to include procedures for preparing, evaluating, and ratifying the Village Forest management plan. Planned activities that can be implemented as part the business permit work plan and in line with the Padang Tikar Landscape forest community include: 1) forest conservation, protection, and monitoring that specify the conservation of flora and fauna, the prevention of illegal logging, and the prevention of forest fires; 2) utilization of the forestry area, including the utilization of timber forest products, agroforestry, silvopasture, and silvofishery; and 3) the utilization of environmental services, both through carbon storage and sequestration and institutional development.

MoEF Regulation No. P.70/MENLHK/SETJEN/KUM.1/12/2017 on Procedures for the Implementation of REDD, the Role of Conservation, Sustainable Management of Forests, and Enhancement of Forest Carbon Stocks

This regulation functions as a guideline that aims to achieve the implementation of REDD+, following the requirements of the Conference of the Parties (COP) UNFCCC on REDD+ and remaining consistent with national policies, along with encouraging REDD+ actors to move towards the full implementation of REDD+ (e.g., results-based payments) to support the achievement of targets for implementing nationally determined contributions (NDCs) in the forestry sector. The implementation of REDD+ represents part of the effort to reduce emissions resulting from deforestation and forest degradation, to conserve and enhance of forest carbon stocks, and to implement sustainable forest management. REDD is conducted gradually, with the goal of completing it through provision of results-based payments.

Decree of the Governor of West Kalimantan No. 770/DPRKPLH/2017 on the Establishment of REDD Groups (Pokja).

The principal responsibility of the Pokja group is to measure and report GHG emissions. Furthermore, the West Kalimantan REDD+ Pokja group is responsible for establishing alliances and institutions and undertaking verifications to adapt to and mitigate the impacts of climate change. In the end, its most important duty is to build a database, along with its resulting publication.

Regional Regulation of West Kalimantan Province No. 6 of 2018 on Sustainable Land-Based Business Management

With an area of 14 million ha, the province of West Kalimantan acts as the primary basis for those sectors of the community's economy which rely on forestry, agriculture, plantation, and mining. To reduce the rate of deforestation and forest degradation in West Kalimantan, policies on sustainable land-based business management are needed. Land conservation in West Kalimantan currently has two legal umbrellas, namely Regional Regulation No. 6 of 2018 and Governor Regulation No. 60 of 2019. The main purpose of the issuance of the regional regulation is to initiate the conservation of land, reserving around seven percent of the region's total area for land-based businesses and activities, such as plantations, agriculture, mining, and forestry. The conservation area is expected to provide economic benefits for the community, balanced with the need for conservation and sustainable production. In addition, the company must also ensure that the areas that have been identified as being of HCV and HCS are managed according to the principles of conservation and sustainable production, and that this involves local communities from across the concession.

Regional Action Plan for Reducing Greenhouse Gas Emissions (RAD-GRK) in West Kalimantan Province No. 125 of 2020.

The scope of the action plan includes three sectors, such as land (e.g., forestry, peatland, agriculture, plantation, and farm), energy (e.g., transportation, energy supply, and use), and waste management. For the AFOLU sectors, mitigation action plans covering forestry and peatland in main activities include: the prevention of deforestation, degradation, and conversion of swamp forest areas/secondary dryland forest in non-forest estate/*area penggunaan lain* (APL) lands; the prevention of deforestation, degradation, and conversion of swamp forest/secondary dryland forest in peatland APL; the prevention of deforestation, degradation, and conversion of swamp forest/secondary dryland forest in a protected forest; the prevention of deforestation, degradation, and conversion of secondary swamp forest in peat protected forest; the development of community gardens in less productive areas; and the planting of critical lands in APL. Cumulative emissions in 2011–2030 under a business-as-usual scenario are predicted to reach 73,614,729.48 tCO_{2e}, while cumulative emission projections in 2011–2030 after implementation of the mitigation actions reach 679,355,670.80 tCO_{2e}.

MoEF Regulation No. P.30 of 2009 Concerning the Implementation of REDD Procedures

In order to improve forest management, REDD activities aim to prevent and reduce emissions from deforestation and forest degradation. The goal of a REDD project is to prevent deforestation and forest degradation so as to promote sustainable forest management and improve people's well-being. Forest management operations within the scope of REDD implementation include determining the reference emission level before REDD implementation, monitoring in accordance with the plan, and submitting the monitoring report to the Minister, all of which are responsibilities of the REDD proponent.

Presidential Regulation No. 98 of 2021 on the Implementation of Carbon Economic Value for the Achievement of NDC Target and GHG Emission Control in National Development

Indonesia has committed to achieving an unconditional GHG emission reduction of 29% (i.e., resulting from its own effort) and a conditional reduction of 41% (i.e., with international support) by 2030. In the Presidential Regulation on the carbon economic value (CEV), there are several carbon trading mechanisms that are regulated, namely trade between two business actors through a cap-and-trade scheme, offsetting emissions through a carbon offset scheme, result-based payments, and levies on the carbon tax, along with a combination of existing schema. This carbon pricing regulation is able to support those mitigation and adaptation efforts that have been implemented in alignment with the purpose of the Padang Tikar project, such as controlling forest fires and preventing deforestation and degradation. Article 77 states that for international carbon trading, the ministry manages the cooperation of mutual recognition. This will be done by having mutual information disclosure of MRV standard used, carrying out conformity assessment against international and/or national standards including statement of its result, conducting the mutual recognition cooperation, and registering certifications that are recognized by both parties at SRN. This platform is an important tool based on this regulation, which any emission reduction project must register under SRN in order to get approval on its emission reduction certificate which allows it to be used for trading. The purpose of the SRN, as mentioned in Article 69, paragraph 3, includes: acting as a basis for the government's recognition of the contributions made to achieving NDC targets through the application of CEV; compiling data and information on mitigation actions and CEV resource application; efforts to avoid double counting climate change mitigation actions; and overseeing transitional materials.

The list below shows the Village Forest permit for each village in the Padang Tikar Landscape:

- SK.515/Menlhk-PSKL/PKPS/PSL.0/2/2017 Tentang Pemberian HPHD kepada LPHD Batu Ampar
- SK.516/Menlhk-PSKL/PKPS/PSL.0/2/2017 Tentang Pemberian HPHD kepada LPHD Teluk Nibung
- SK.518/Menlhk-PSKL/PKPS/PSL.0/2/2017 Tentang Pemberian HPHD kepada LPHD Tanjung Harapan
- SK.519/Menlhk-PSKL/PKPS/PSL.0/2/2017 Tentang Pemberian HPHD kepada LPHD Ambarawa
- SK.520/Menlhk-PSKL/PKPS/PSL.0/2/2017 Tentang Pemberian HPHD kepada LPHD Sungai Besar
- SK.521/Menlhk-PSKL/PKPS/PSL.0/2/2017 Tentang Pemberian HPHD kepada LPHD Tasik Malaya
- SK.522/Menlhk-PSKL/PKPS/PSL.0/2/2017 Tentang Pemberian HPHD kepada LPHD Medan Mas

- SK.523/Menlhk-PSKL/PKPS/PSL.0/2/2017 Tentang Pemberian HPHD kepada LPHD Padang Tikar 1
- SK.524/Menlhk-PSKL/PKPS/PSL.0/2/2017 Tentang Pemberian HPHD kepada LPHD Nipah Panjang
- SK.525/Menlhk-PSKL/PKPS/PSL.0/2/2017 Tentang Pemberian HPHD kepada LPHD Sungai Jawi

Both the compliance and voluntary carbon markets will follow Presidential Regulation No. 98 of 2021, while derivative regulations are still being developed by the Indonesian government. This project will follow current and future regulations that apply in the PA.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project does not participate under other GHG programs.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been submitted to or rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes

☒ No

The project neither has, nor intends to generate, any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program. The VCS Program has a central project database, which lists each approved project. The VCS Project Database is the central storehouse of information on all projects validated according to VCS criteria and all Verified Carbon Units (VCUs) issued under the program. Every VCU can be tracked from issuance to retirement in the database, allowing buyers to ensure every credit is real, additional, permanent, independently verified, uniquely numbered, and fully traceable online. This project has not been registered under any other credited activity, and no VCUs have been assigned to the PA thus far. Therefore, any possibility of double counting credits has been eliminated. As mentioned in section 1.14, Presidential Regulation No. 98 is currently in effect to ensure that no double counting can take place and that the project will be registered with the SRN.

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes

☒ No

The project has not sought or receive another form of GHG-related environmental credit, including renewable energy certificates.

1.17 Sustainable Development Contributions

According to data and statistics provided in the 'Sustainable Development Report 2021', Indonesia ranks 97th out of 165 countries in the Sustainable Development Goal (SDG) Index rankings, with an overall score of 66.3 out of 100. This score indicates a country's total progress toward achieving all 17 SDGs. Previously, Indonesia has been awarded scores of 66.04, 66.27, and 66.22 in the 2018, 2019, and 2020 'Sustainable Development Reports' respectively.

Currently, Indonesia has achieved zero SDGs, with nine of the remaining SDGs in particular presenting major challenges to improvement. The trends show that progress towards achieving three major SDGs (i.e., sustainable cities and communities, life below water, and life on land) has either stagnated or is occurring at less than 50% of the required rate (see Figure 6).



Figure 6: Current assessment and trends on SDGs achievement in Indonesia⁴⁴

As the above data shows, Indonesia faces major challenges in achieving its SDGs. This project intends to contribute to at least six SDGs (see Table 7)

⁴⁴ Available at: <https://dashboards.sdindex.org/static/profiles/pdfs/SDR-2021-indonesia.pdf>.

Table 7: Project SDGs Contributions

SDG	Contribution description	Monitoring description
SDG 1: No Poverty	Padang Tikar Landscape aims to employ local people	- Number of people employed - Percentage of employees earning a salary above the poverty headcount ratio at IDR 2,467,630 per month⁴⁵
SDG 4: Quality Education	Padang Tikar Landscape provides training and workshops related to alternative livelihood activities, climate mitigation activities, and biodiversity conservation and enhancement	- Number of training sessions and workshops - Numbers of female/male participants in training sessions and workshops
SDG 8: Decent Work and Economic Growth	Padang Tikar Landscape creates various alternative livelihood activities, such as crab production, honey production, and coconut charcoal production	- Number of people working related to alternative livelihood activities - Generated or monthly incomes related to alternative livelihood activities
SDG 13: Climate Action	Integrate climate change measures into strategies and planning	Net estimated GHG emission reductions in the PA, measured against the baseline scenario
SDG 14: Life Below Water	Padang Tikar Landscape aims to increase the abundance, diversity, and critical habitat characteristics of endangered and endemic wildlife	Number of fauna species below water found in the PA in each verification period
SDG 15: Life on Land	Padang Tikar Landscape aims to increase the abundance, diversity, and critical habitat characteristics of endangered and endemic wildlife	- The area of protected forests in the Padang Tikar Landscape (ha) - Number of fauna and flora species found on land in the PA in each verification period

1.18 Additional Information Relevant to the Project

⁴⁵ Governor of West Kalimantan Decree No.1468/DISNAKERTRANS/2021.

Leakage Management

Considering that the project activities will be focused on reducing GHG emissions from unplanned deforestation, the project proponents ensure that project activities will have a positive impact on surrounding areas. The following project activities may provide strategies for successfully mitigating leakages:

- 1) increasing the ability of the forest patrol to protect and safeguard forests and biodiversity;
- 2) restoring critical mangrove and peat ecosystems;
- 3) strengthening the community's sustainable livelihood business model; and
- 4) increasing community capacity through training, comparative studies, and internships.

By means of the project monitoring activities, satellite imaging, and social and government cooperation with project monitoring and its surrounding area, the project proponents believe that the success of the business plan will generate an increased number of sustainably managed areas with project activities. Further details can be found in sections 1.11 and 4.3.

Commercially Sensitive Information

Examples of commercially sensitive information not made publicly available include:

- 1) financial plans, budgets, and projection, including details of the organization's project implementation, work plans, and initial and long-term project budgets over the course of the project (i.e., 30 years' worth of projections);
 - 2) original data, electronic form files, computer model codes, and confidential satellite imagery, including GIS project boundaries, land use, and remote sensing data, biomass inventories, hydrological assessment models, carbon accounting, and social assessments in the Padang Tikar Landscape PA; and
 - 3) agreements and contracts between project proponents, other project entities, and stakeholder groups, particularly regarding the implementation of project activities.
- Specific information is personally available to project validators upon request.

However, information related to the determination of the baseline scenario, the demonstration of additionality, and the estimation and monitoring of GHG emission reductions and removals, including operational and capital expenditures, is not considered to be commercially sensitive and must be provided in public versions of the project documents.

Further Information

No further information to disclose.

2 SAFEGUARDS

2.1 No Net Harm

The project activities do not cause any potential negative socioeconomic impacts to stakeholders. The project does not result in net negative impacts on the well-being of other stakeholder groups. A communication channel or grievance mechanism has been discussed and established for stakeholders to express their concerns and to solve potential conflicts and grievances that may arise during project planning and implementation.

During stakeholder consultations, non-technical information, forest management, forest inventory, community development, and project benefits have been presented to stakeholders to increase their understanding of climate change mitigation, the importance of forestry, and the positive impacts of carbon project development. Therefore, project activity implementation is not expected to result in any potentially negative impacts.

2.2 Local Stakeholder Consultation

Stakeholder Consultations

Consultation has been essential in accommodating and securing the involvement of all stakeholders in the process of developing emission reduction projects in the Padang Tikar Landscape. Prior to holding consultations in each stakeholder group, the project owner carried out a series of field visits and initial outreach to community groups. The field visits and outreach were very helpful in the process of identifying stakeholders who wished to be involved in discussions and provide inputs related to carbon development projects.

Meetings (e.g., workshops, conversations, and informative talks) have been arranged with the different audiences with the purpose of informing them about the management of the project and the difficulties of its implementation. These spaces will also serve to provide information on the steps to be followed, the programmed activities, and the functions, duties, and rights of each of the parties involved.

The subject of these meetings has ranged from general issues, such as project activities and the work plan, the publication of monitoring reports, and general information on the project's legal basis, to very specific issues, such as individual village participatory plans, fire management, and agroforestry development. In addition to informational meetings at the community level, the project has conducted and participated in many stakeholder meetings at the national, provincial, and district levels, providing information on both the ongoing operation of the project and future plans for the project while disseminating information on the carbon project concept.

Listed below are selected insights from the community included in the project design:

1) Participatory and collaboration plans

To ensure collaborative cooperation within village community groups, the community suggested actively involving all layers of village government, including the village council (BPD), besides the LPHD and the head of the village. The involvement of the BPD is crucial for ensuring that the benefits generated from carbon asset development projects are able to be distributed evenly for the activities that prevent deforestation and forest degradation.

2) PA and activities

During the consultation stages, communities were involved in participatory mapping for the PA, providing information related to the project boundaries and relevant information on activities that are suitable for implementation at village level.

Community Costs, Risks, and Benefits

Since the project started in 2017, project partners have established possible conservation efforts within the Padang Tikar Landscape related to both consultation and socialization. This socialization resulted in the identification of activities that will be beneficial to conservation efforts, such as fire and logging patrols, canal blocking and peat depth measurements as rewetting efforts, and alternative livelihoods that will reduce the tendency of forest encroachment.

As time has passed in the PA, project partners have further developed consultation and socialization specifically to discuss the potential risk and benefits of carbon asset development within the PA. Several risks identified included the ability of community groups to implement the required activities, which then led to the necessity of assistance from project partners in the field, limited data collection related to the baselining of biodiversity potential, and the possible adjustment of conservation efforts in Batu Ampar to facilitate local stakeholder well-being.

The identified risks are already being appropriately discussed with local stakeholders in the Padang Tikar Landscape, encouraging community participation with the project design and its related activities. After following discussions related to the potential risks and costs of the project, the community has gained enough knowledge to make a sound and consensual decision regarding a project design and activities that will contribute to emission reductions while fulfilling the needs of the community.

The project will maintain active communication channels with all stakeholders throughout the crediting period. This will allow the community to voice all concerns, comments, questions, requests, and complaints related to the project.

Figure 7 below includes photographs documenting a consultation meeting held in Padang Tikar Landscape.



Figure 7: Photographs of local stakeholder consultation

Relevant Laws and Regulations Related to Worker's Rights

The project in the Padang Tikar Landscape complies with national and provincial laws and regulations related to worker's right, such as:

- 1) Law No. 13/2003, ensuring worker safety and the management of manpower. Law No. 13/2003 also stipulated the safety regulations that workers in conservation projects should be provided with;
- 2) The Manpower and Transmigration Regulation No. 233/2003, explaining the types of work that can be carried out continuously. This regulation also explains the exemption of working hours for certain sectors, including conservation; and
- 3) The Manpower and Transmigration Regulation Decree No. 100/MEN/IV/2014, which explains the implementation of definite terms of employment.

Information to Stakeholders on the Validation and Verification Process

Information regarding project development, including validation and verification processes, has been communicated to all stakeholders across both the engagement meeting and the Local Stakeholder Consultation (LSC) meeting. All stakeholders were well informed about the validation and verification process, in terms of the period, method, and manner in which validation and verification will be carried out.

During the socialization and consultation process, key project information was disseminated in a short summary, designed to ease understanding of the project for the stakeholders. Short infographics related to project development, key information, and the project summary were distributed to local stakeholders, especially to those who reside close to the PA. As the aim of these infographics was to aid stakeholders with understanding the project, they have been provided in Indonesian.

In addition to both the consultation and the infographics, notification of the validation and verification process, including details on how to comment on the project, will also be issued to all stakeholders.

Site Visit Information and Opportunities to Communicate with the Auditor

The project will arrange all measures necessary for conducting site visits in an orderly manner as part of the validation and verification process, including meetings with all relevant stakeholders.

2.3 Environmental Impact

Environment impact assessments will operate at two different levels. First, the integrity of forest cover in the PA and its connectivity with surrounding forest areas, such as the leakage belt forest area, will be assessed. Second, the biodiversity of fauna and flora species, both below water and on land, will be assessed, focusing on specific groups such as mammals, birds, reptiles, amphibians, and fish. The monitoring parameters will be specifically listed in the Climate, Community, and Biodiversity project design document and will be related to outputs, outcomes, and impacts. The frequency of assessment will be monitored every 3–5 years.

2.4 Public Comments

This is a preliminary version of the document. The public comment period has not yet been completed.

2.5 AFOLU-Specific Safeguards

Stakeholder Identification Process and Description

This project has been started as a follow-up to a previous round of stakeholder engagement held in the Landscape. The previous engagement was focused on supporting the villagers with

acquiring the 'Village Forest' permit through the MoEF's social forestry scheme. The permit is seen as an opportunity for developing the community by improving livelihoods with a sustainable approach while also achieving the protection of the forest, as mandated by the Indonesian government in the terms of the permit.

The mandate of the Village Forest permit is in line with the project's objective. The village community, as the owners of the Village Forest rights, have remained engaged via the livelihood approach throughout the project lifetime, participating as part of a strategy to slow down the deforestation rate and maintain the ecosystem function of the area.

The stakeholder analysis was started by mapping and defining the PA and project zone within the Padang Tikar Landscape. The 'PA' is defined as an area inside the Village Forest eligible for emission reduction calculations, while the 'project zone' is defined as an adjunct region to the PA where the community live and have been engaged for project activities that focus on livelihood and community development. Once the area had been clearly defined, the analysis was continued to identify which individuals, groups, or other stakeholders would be directly or indirectly affected by the project. This analysis also focused on identifying which stakeholders would have an impact, whether positive or negative, on the project objectives.

From the analysis, stakeholders in the Landscape were divided into two main groups, i.e., communities and other stakeholders. 'Communities' represent individual and/or community groups that are directly and indirectly affected by the project, while 'other stakeholders' represent the groups that are indirectly involved in supporting the project objective.

Listed below are all the communities, community groups, and other stakeholders in the project, alongside a brief description of their rights, interests, and relevance to the project activities (Table 8). Each of these stakeholders were invited to participate in the LSC sessions.

Table 8: Stakeholder identification and description

Group	Stakeholder	Rights	Interests	Relevance
Communities	Local communities (villagers)	- Owner of the village forest management rights - Receive training and assistance with economic development. Responsible for managing the forest responsibly and sustainably under	- Utilize the Village Forest area as a business area - Participate in the development of strategies for the management of the Village Forest	Very high on climate, livelihood, and biodiversity objectives

Group	Stakeholder	Rights	Interests	Relevance
		the mandate of the Village Forest permit		
	Community leaders	● Provide direction and guidance to the community ● Receive assistance with supporting the development of Village Forest programs	● Participate in the development of strategies for Village Forest management ● Help oversee implementation of the Village Forest program	Low on livelihood aspects
	Fire starters	● Participate in the development of strategies for Village Forest management ● Obtain proper fire prevention and mitigation training	● Participate in the development of strategies for Village Forest management ● Participate in the development of a program for land clearing without burning	Very high on climate, livelihood, and biodiversity objectives
	Illegal loggers	● Participate in the development of strategies for Village Forest management ● Obtain proper training in mitigating illegal logging ● Participate in economic activities which mitigate illegal logging	● Participate in the development of strategies for Village Forest management ● Participate in economic developments in the Village Forest which prevent illegal logging	Very high on climate, livelihood, and biodiversity objectives
	Palm oil smallholders	● Participate in the development of strategies for Village Forest management ● Receive training and assistance with economic activities that are more	● Utilize the Village Forest area as a business area ● Participate in the development of strategies for Village Forest management	Very high on climate and biodiversity objectives

Group	Stakeholder	Rights	Interests	Relevance
		sustainable than palm oil production		
	Fishermen	● Utilize the Village Forest area (mangrove forest) as a fishing ground for crabs, shrimp, etc. ● Participate in the development of strategies for Village Forest management ● Receive training and assistance with the development of fishery products	● Participate in the development of strategies for Village Forest management ● Participate in sustainable business/economic development for the mangrove forest and fishery sectors	Very high on climate, livelihood, and biodiversity objectives
	Farmers	● Participate in the development of strategies for Village Forest management ● Receive training and assistance with economic development in the agricultural sector	● Utilize the Village Forest area as a business area ● Participate in the development of strategies for Village Forest management	Very high on livelihood aspects
	Middlemen	● Generate sustainable revenue from new business activities developed in the villages ● Obtain assistance with the development of fair fishery products	Participate in business development in the Village Forest	Low on livelihood aspects
Community groups	Village government	● Owner of Village Forest management rights ● Exercise Village Forest surveillance rights, including changing the LPHD structure and	● Coordinate between the community and the LPHD ● Community economic development in the village,	Medium on livelihood aspects

Group	Stakeholder	Rights	Interests	Relevance
		ensuring Village Forest management is functioning as planned • Take the lead in defining the direction of the Village Forest and in planning and developing strategies for its management • Obtain training and assistance with Village Forest development	including in the Village Forest • Prevent deforestation and wildfires from occurring in the Village Forest	
	LPHD	• Hold a mandate with the MoEF to manage and lead Village Forest management • Take the lead in defining the direction of the Village Forest and in planning and developing strategies for its management • Implement all existing programs in village forests (e.g., ecosystem services, timber products, and non-timber forest products) • Utilize the Village Forest area as a business area • Obtain training and assistance with	• Coordination between the village government and community • Community economic development in the village and Village Forest • Prevent deforestation and wildfires from occurring in the Village Forest • Implement the Village Forest program outlined in the annual work plan (RKT) and village forest management plan (RPHD)	Very high on climate, livelihood, and biodiversity objectives

Group	Stakeholder	Rights	Interests	Relevance
		Village Forest management		
	KUPS	• Develop existing businesses and potential commodities in Village Forest areas • Utilize the Village Forest area as a business area • Participate in the development of strategies for Village Forest management • Obtain training and assistance with economic development	• Utilize the Village Forest area as a business area • Participate in the development of strategies for Village Forest management	Very high on climate, livelihood, and biodiversity objectives
	Fishermen groups	• Utilize the Village Forest area (mangrove forest) as a fishing ground for crabs, shrimp, etc. • Participate in the development of strategies for Village Forest management • Receive training and assistance with the development of fishery products	• Participate in the development of strategies for Village Forest management • Participate in sustainable business/economic development in the mangrove forest and fishery sectors	Very high on climate, livelihood, and biodiversity objectives
	Forest farmer groups	• Participate in the development of strategies for Village Forest management • Receive training and assistance with economic development in the non-timber forest product sector	• Utilize the Village Forest area as a business area • Participate in the development of strategies for Village Forest management	Very high on livelihood aspects

Group	Stakeholder	Rights	Interests	Relevance
	Community fire care groups/Masyarakat Peduli Api	• Conduct forest security and protection • Obtain training in and assistance with the prevention and control of forest and land fires	• Participate in the development of strategies for Village Forest management • Participate in the prevention and control of forest and land fires	Very high on climate and biodiversity objectives
Other stakeholders	Private sector	Support the program by providing business development capital and/or participating in the marketing and development of products	• Ensure that illegal logging and wildfires are not occurring in the concession area • Participate in the economic development of the Village Forest	Medium on climate, biodiversity, and livelihood objectives
	LPHD	• Monitor the implementation of the Village Forest permit • Empower communities surrounding the Village Forest • Coordinate with village governments and the LPHD for Village Forest management • Run the program in the Village Forest	• Develop all programs (protection and economy) in the Village Forest • Ensure that there is no illegal logging or forest fire in the area • Ensure that the LPHD institution is able to sustain 35 years of management	Very high on climate, livelihood, and biodiversity objectives
	Other government institutions	Provide suggestions and input to ensure the program is in accordance with the targets of the West	Obtain project activity reports for integration, learning, and input from government planning	Medium to high on climate, biodiversity, and

Group	Stakeholder	Rights	Interests	Relevance
		Kalimantan provincial government		livelihood objectives
	Other non-governmental organizations (NGOs)	Provide suggestions and input (if needed) for project development and participate in relevant aspects of the project (if needed)	Obtain project progress information to use as a model for the replication and evaluation of other programs	Medium on climate, livelihood, and biodiversity objectives
	Village facilitators	Provide training, knowledge, and community guidance at the village level	● Run the program planned by the Ministry of Villages ● Assist the village government in carrying out the program ● Collaborate on economic development in the village	Low on livelihood aspects

Risks to Stakeholders Due to Project Implementation and Risk Mitigation

The project activities do not expect to bring risk to the identified stakeholders, to their resources, or to their ownership rights. On the contrary, the project was designed to promote positive impacts on villagers and communities in the Padang Tikar Landscape and its surrounding areas. However, risks were discussed with identified stakeholders during the LSC as part of the doubts expressed by communities. In this sense, the project proponent addressed issues regarding benefit sharing (i.e., share and mechanism) and activities related to mangrove areas.

In response to these doubts, the team has conveyed that the communities' livelihoods depend on the deforestation of small patches for agriculture and charcoal production for household use, and that the project should not interfere with this local dynamic, also attested to communities as a mitigation measure. These messages are understood by the communities. The conclusion reached is that, in the elaboration of participatory project design workshops, the relevant communities must consent to the execution of the project before project activities

commence. In this way, through the stakeholder consultation process and social engagement protocols, the risks perceived by the community are mitigated.

Free, Prior, and Informed Consent

The Padang Tikar Landscape project adopts free, prior, and informed consent principles throughout the community consultation process. As the mangrove area is important to community livelihoods, an agreement between the project proponent and the community representative for the Village Forest (i.e., the LPHD) in 10 villages, mandated by the MoEF, has been obtained to ensure the development of the project within the Village Forest area. The mandates stated in the Village Forest permit letter include: the responsibility of the LPHD for protecting their area from environmental destruction and pollution; maintaining the forest function; implementing forest protection; and carrying out forest planting and maintenance. Those responsibilities are in line with the aim of this project. In order to comply with these responsibilities, the local community and the LPHD require support and assistance from other parties with expertise in sustainable forest management. Along with the issuance of the Village Forest permit, the LPHD requested that the MoEF recommend a local NGO to fill the role of supporting institution. Therefore, project proponents aim to build the LPHD's capacity for achieving their responsibilities as outlined in the Village Forest permit letter.

Community engagement has become an important aspect of this project for the project proponent and will be continued throughout the project lifetime.

Feedback and Grievance Redress Procedure

Processes for ensuring ongoing communication and consultation with local stakeholders, including the establishment of a grievance redress procedure to receive and resolve any complaints and conflicts which may arise between the project proponent and local stakeholders, has been designed during the second phase of the LSC, as shown in Figure 8.

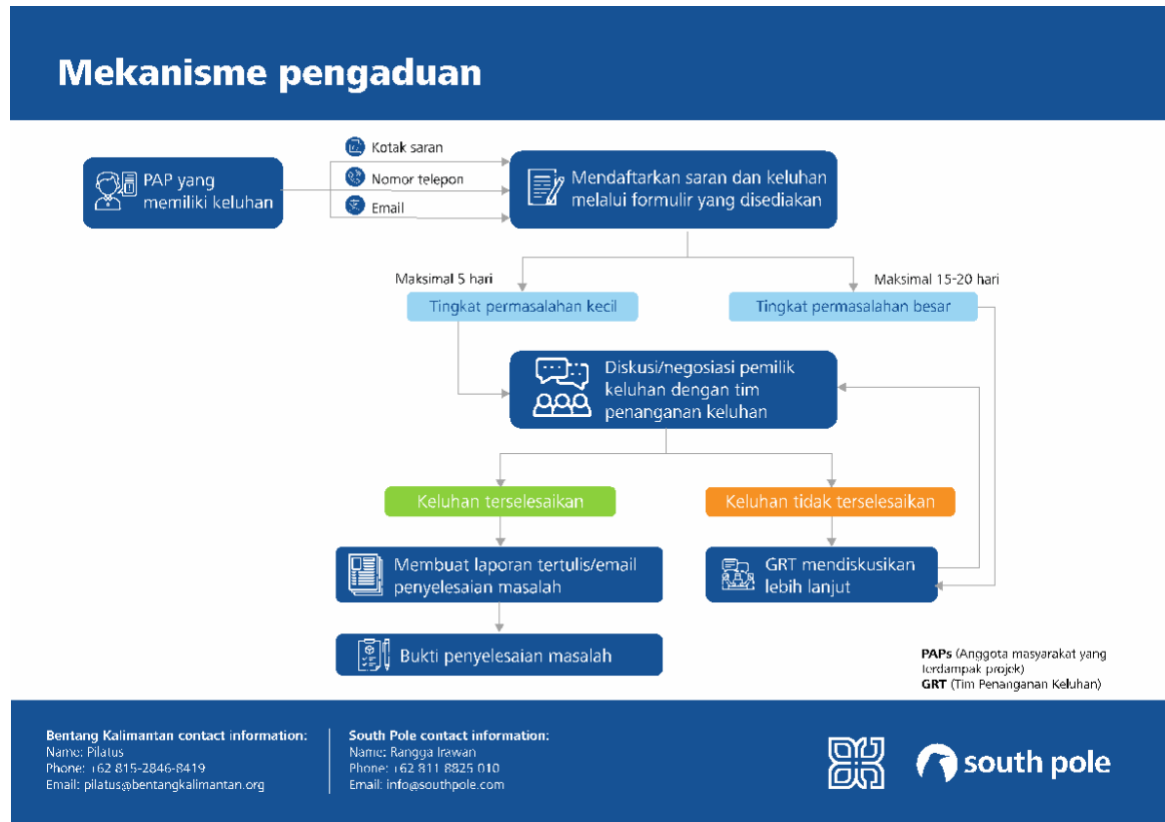


Figure 8: Grievance redress procedure in Padang Tikar Landscape

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The list below references the methodology, modules, and tools used in the project scope:

- 1) VM0007 REDD+ Methodology Framework (REDD+MF), Version 1.6
- 2) VMD0001 Estimation of carbon stocks in the above- and below-ground biomass in live tree and non-tree pools (CP-AB), Version 1.1
- 3) VMD0004 Estimation of carbon stocks in the soil organic carbon pool (CP-S), Version 1.0
- 4) VMD0007 Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation (BL-UP), Version 3.3
- 5) VMD0010 Estimation of emissions from activity shifting for avoiding unplanned deforestation and avoiding unplanned wetland degradation (LK-ASU), Version 1.2

- 6) VMD0013 Estimation of greenhouse gas emissions from biomass and peat burning (E-BPB), Version 1.2
- 7) VMD0044 Estimation of emissions from ecological leakage (LK-ECO), Version 1.1
- 8) VMD0052 Tool for the Demonstration of additionality of tidal wetland restoration and conservation project activities (ADD-AM), Version 1.0
- 9) VT0001 Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forest and Other Land Use (AFOLU) Project Activities, Version 3.0
- 10) AFOLU Non-permanence Risk Tool, Version 4.0

3.2 Applicability of Methodology

This methodology provides a set of methodological modules for REDD and Wetland Rewetting and Conservation (WRC). The modules, when used together, quantify reductions and removals due to avoided deforestation. This methodology is applicable to forests, forested wetlands, and forested peatlands susceptible to deforestation or degradation in the absence of project activities.

Considering that the general scope of the methodology is tailored to the conditions and characteristics of the project, the applicability conditions are then evaluated to determine their potential use in certifying associated mitigation results and emission reductions expected following the implementation of project activities.

Considering the sectoral scope and project type of the Blue Carbon Project, the applicability of the methodology will be evaluated by:

- 1) reduction of emissions from AUDD; and
- 2) unplanned deforestation of intact or partially drained wetlands (AUWD).

Table 9: Analysis of applicability conditions for the VM0007 REDD+ Methodology Framework

Type of activity	Application	Justification
All types of activity	All land areas registered under the Clean Development Mechanism or any other GHG program, whether voluntary or compliance-oriented, must be transparently reported and excluded from the PA. The exclusion of land in the PA from any other GHG program must be monitored over	There are no other GHG programs registered for the PA.

Type of activity	Application	Justification
	time and detailed in the monitoring reports.	
	Land in the PA must qualify as a 'forest' for at least 10 years prior to the project start date, as per the definition used by the VCS. Mangrove forests are excluded from any tree height requirement in this definition as they consist of close to 100% mangrove species, which often do not reach the same height as other tree species and occupy contiguous areas. Likewise, their functioning as a forest is independent of tree height.	The definition of 'forest' for Indonesia is land spanning for more than 0.25 ha, dominated by trees higher than five m at maturity, and with canopy cover of more than 30%. Based on this definition, and as a result of the eligibility analysis, ⁴⁶ it was found that 58,765 ha has been forest for at least 10 years (i.e., the period 2006–2016) prior to the project start date. Mangroves were excluded from any height requirement established in Indonesia's forest definition.
	If land within the PA is peatland or tidal wetland and emissions from the soil carbon pool are deemed significant, the relevant WRC modules must be applied alongside other relevant modules.	As the project is developed in coastal wetlands, the relevant WRC modules were applied (VMD0052), according to the methodology, for the estimation of emission reductions.
	Baseline deforestation and forest degradation in the PA fall within one or more of the following categories: • unplanned deforestation (VCS category AUDD); • planned deforestation/degradation (VCS category 'avoiding planned deforestation'); and • degradation through extraction of wood for fuel (e.g., fuelwood and charcoal production) (VCS category AUDD).	The processes found in the area correspond to AUDD.

⁴⁶ The supporting documents are available in the folder Section 3/Eligibility Analysis.

Type of activity	Application	Justification
	Leakage avoidance activities must not include: • agricultural lands that are flooded to increase production (e.g., paddy rice); or • intensifying livestock production through the use of feedouts and/or manure lagoons.	These options were not included as leakage management activities.
AUDD	Baseline agents of deforestation must: 1) clear the land for settlements, crop production (agriculturalist), or ranching, where such clearing for crop production or ranching does not amount to large-scale industrial agriculture activities; 2) have no documented and uncontested legal right to deforest the land for these purposes; and 3) be either residents in the Reference Region for Deforestation (RRD) or immigrants. This methodology must not be used under any other condition.	The agents of deforestation in the forest areas are residents of the RRD and are led by small-scale crop production and charcoal production.
	If, in the baseline scenario of avoiding unplanned deforestation project activities, post- deforestation land use constitutes reforestation, this methodology may not be used.	The post-deforestation activities, under the baseline scenario, do not include reforestation.
All WRC activity	WRC activities are not eligible under the following conditions: • project activities lower the water table, unless the project converts open water to tidal wetlands, or improves the hydrological connection to impounded waters; • changes in hydrology do not result in the accumulation or maintenance of soil carbon stock;	Activities related to wetland conservation in the PA: • do not lower the water level; • do not change in hydrological connectivity; • do not generate an increase in GHG emissions outside the PA;

Type of activity	Application	Justification
	- hydrological connectivity of the PA with adjacent areas leads to a significant increase in GHG emissions outside the PA; or - project activities include the burning of organic soil. Nitrogen fertilizer(s), such as chemical fertilizer or manure, are applied in the PA during the project crediting period.	- do not include organic soil burning activities; and - do not consider the application of nitrogen fertilizers.
CIW	This methodology is applicable to conservation of undrained and partially drained tidal wetland activities in PAs that meet the VCS definition for peatland.	The PA meets the definition of peatland proposed by the VCS in methodology VM0007, as forests correspond to mangrove and peatland forests. The specific project activity focuses on the conservation of wetlands at risk from unplanned deforestation, an activity eligible under the methodology.
AUWD	Baseline agents of wetland degradation: 1) cause an alteration in the hydrology of the PA, either involving drainage, interrupted sediment supply, or both, and/or a loss of soil organic carbon; 2) have no documented and uncontested legal right to degrade the wetland; and 3) are either residents in the reference region for wetland degradation or immigrants. This methodology must not be used under any other condition.	Wetland degradation agents in the baseline scenario cause the loss of soil organic carbon due to unplanned deforestation. The scenario has no documented and uncontested legal right to degrade wetlands.

3.3 Project Boundary

Spatial Boundaries

Below is a description of the spatial boundaries of the project, considering three types of area used in the methodology. As the project is in the AUDD category, the PA, the reference region for projecting rate of deforestation, the reference region for projecting location of deforestation, and the leakage belt (LK) area are therefore considered. Figure 9 presents a map with applicable boundaries.

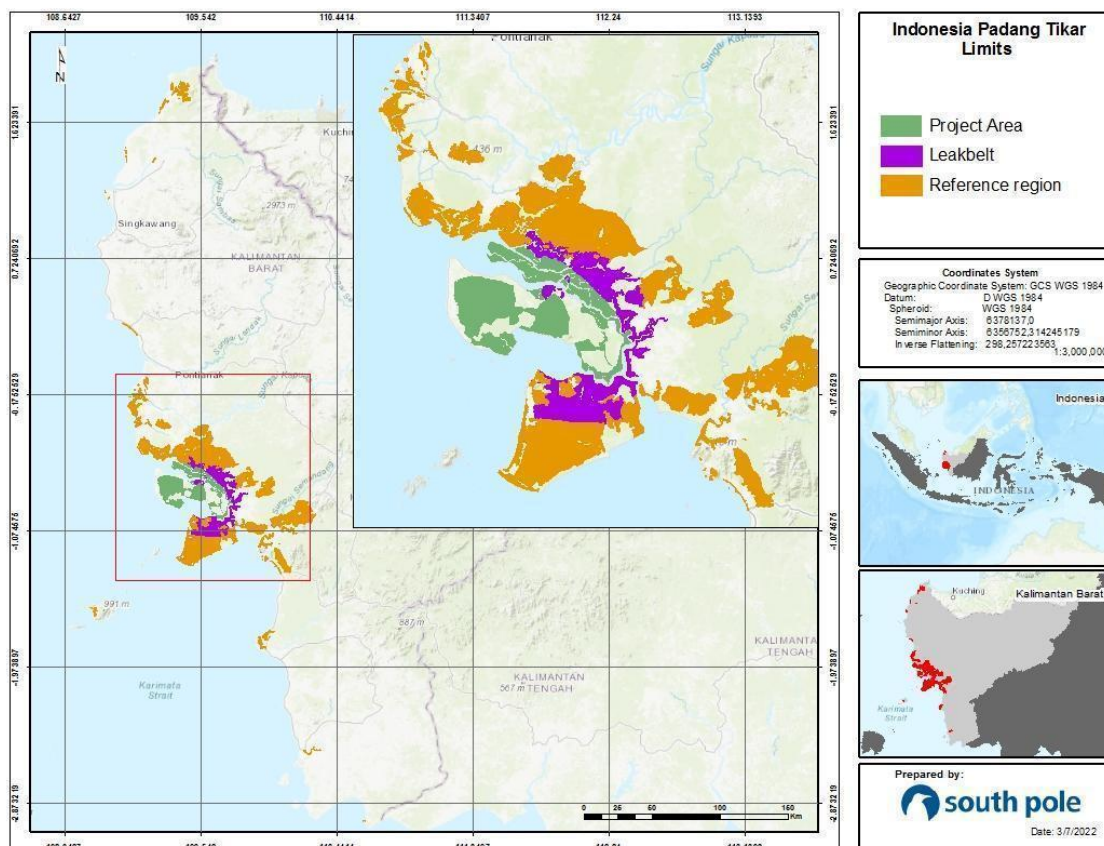


Figure 9: Spatial boundaries

Project Area

According to VMD0007 (BL-UP module), the PA is the discrete parcels of land, under threat of deforestation, on which the project proponents will undertake activities to reduce unplanned deforestation. The project must be 100% forested at the project start date (e.g., 2017).

In this case, the PA corresponds to secondary dryland forest, secondary swamp forest, and secondary mangrove forest within the Padang Tikar Landscape that have remained as forests throughout the period 2006–2016. The PA covers 58,672.7 ha and is located between 0°37'2.11" to 0° 0'13.18" South latitude and 109°15'26.49" to 109°44'53.91" East longitude. The PA does not include forests in which REDD project activities will not be carried out or which have not been included in the baseline assessment.

For the PA, the only areas considered were those where the project developers will promote activities and where the forest criteria are met for the end of the historical series analyzed, i.e., with stable forests in year zero, or project start date.

Reference Region for Projecting Rate of Deforestation

The reference region for projecting the rate of deforestation (RRD), according to VMD0007 (BL-UP module), does not need to be contiguous with and must not encompass the PA and the LK. The RRD can be composed of several parcels that do not have to be contiguous. Furthermore, the RRD areas must be forested at the start of the historical reference period (e.g., 2006).

The minimum size of the RRD must be calculated as follows:

$$\text{MREF} = \text{RAF} * \text{PA}$$

$$\text{RAF} = 7500 * \text{PA} - 0.7$$

If the RAF, as calculated using the equation above, is less than 1, the RAF must be made equal to 1.

Where:

MREF The minimum size of the reference region for projecting rate of deforestation (ha)

PA Unplanned deforestation PA (ha). In this case, the PA is equal to 58,672.7 ha

RAF Reference Area Factor, or the factor to multiply with the PA to get the minimum reference area; dimensionless

Reference Region for Projecting Location of Deforestation

The reference region for projecting location of deforestation (RRL) must be a single parcel, contiguous with and including the PA and the LK. The RRL must consist of a minimum of 5% non-forest and a minimum 50% forest at the start of the historical reference period (e.g., 2006). The area of forest in the RRL must be equal to the PA of the RRD ($\pm 25\%$). The RRL is only required where location analysis is required or elected.

Leakage Belt

The LK area must be the forest area closet to the PA meeting the minimum area requirement. All parts of LK must, at a minimum, be accessible and reachable by the project baseline deforestation agents, with considerations to agent mobility. The minimum LK area must be equal to at least 90% of the PA.

Temporal Boundaries

- 1) Start and end dates of the historical reference period

In the projection of deforestation under the historic approach, the 'reference period' corresponds to the years that have passed between eight spatial data points. Considering the availability of satellite images for the classification and delimitation of forest areas, the selected historical reference period in this instance was 2006–2009–2011–2012–2013–2014–2015–2016, or the 10 years prior to the project start date.

2) Start and end dates of the project crediting period

From 30-August-2017 to 29-August-2047. The length of the crediting period is 30 years.

3) Duration of the monitoring periods

- I. Baseline reassessment: as per the VCS requirements,⁴⁷ it is mandatory to reassess the baseline every six years for the duration of the project.
- II. First anticipated monitoring period: the first anticipated monitoring period will be from 30-August-2017 to 31-December-2021.
- III. Anticipated subsequent monitoring periods: the subsequent monitoring periods are anticipated to occur in three-year intervals.⁴⁸

The GHG emission sources included or excluded from the REDD project limits are shown in Table 10, while the sources associated with the WRC activity are shown in Table 11.

Table 10: GHG sources from REDD

Source		Gas	Included?	Justification/explanation
Baseline	Burning of woody biomass	CO ₂	Included	The project does not consider biomass burning under the baseline scenario. However, if emissions existed, decreases in carbon stocks due to burning would be counted as a change in carbon stocks due to unplanned deforestation.
		CH ₄	Excluded	Omitted to be conservative
		N ₂ O	Excluded	Omitted to be conservative
		Other	Excluded	No other GHG gases are considered in this project activity
	Combustion	CO ₂	Excluded	Omitted to be conservative

⁴⁷ Refer to paragraph 3.2.7 under the VCS Standard, v4.2, dated January 20, 2022. For all AUDD, APDD (where the agent is unknown), AUC, and AUWD project types, the project proponent shall, for the duration of the project, reassess the baseline every six years and have this validated at the same time as the subsequent verification.

⁴⁸ Potential periods subject to change.

Source		Gas	Included?	Justification/explanation
		CH ₄	Excluded	Potential emissions are negligible
		N ₂ O	Excluded	Potential emissions are negligible
		Other	Excluded	No other GHG gases are considered in this project activity
	Use of fertilizer	CO ₂	Excluded	Potential emissions are negligible
		CH ₄	Excluded	Potential emissions are negligible
		N ₂ O	Excluded	Omitted to be conservative
		Other	Excluded	No other GHG gases are considered in this project activity
Project	Burning of woody biomass	CO ₂	Excluded	Avoided biomass burning is conservatively excluded. However, carbon stock decreased due to burning is accounted as a change in carbon stock if a fire occurs.
		CH ₄	Included	It is conservative to exclude non-CO ₂ gases emitted from woody biomass burning in the baseline, but these must be included if a fire occurs in the PA.
		N ₂ O	Included	It is conservative to exclude non-CO ₂ gases emitted from woody biomass burning in the baseline, but these must be included if a fire occurs in the PA.
		Other	Excluded	No other GHG gases are considered in this project activity
	Combustion	CO ₂	Excluded	The project did not consider the burning of fossil fuels under the baseline scenario
		CH ₄	Excluded	Potential emissions are negligible
		N ₂ O	Excluded	Potential emissions are negligible
		Other	Excluded	No other GHG gases are considered in this project activity
	Use of fertilizer	CO ₂	Excluded	Potential emissions are negligible
		CH ₄	Excluded	Potential emissions are negligible
		N ₂ O	Excluded	The project did not consider the use of fertilizer under the baseline scenario or

Source	Gas	Included?	Justification/explanation
			as a mechanism for preventing leakage. Thus, it was not considered under the project scenario.
	Other	Excluded	No other GHG gases are considered in this project activity

Table 11: GHG sources from WRC

Source		Gas	Included?	Justification/explanation
Baseline	Oxidation of drained peat	CO ₂	Included	Considered under carbon pools
		CH ₄	Included	Required unless <i>de minimis</i> or conservatively omitted
		N ₂ O	Excluded	Excluded, as per the applicability condition in module BL-PEAT
		Other	Excluded	No other GHG gases are considered in this project activity
	Burning of peat or biomass	CO ₂	Excluded	The project does not consider burning soil or biomass under the baseline scenario. According to the E-BPB module, the inclusion of the fire component is always optional in the baseline.
		CH ₄	Excluded	The project does not consider burning soil or biomass under the baseline scenario. According to the E-BPB module, the inclusion of the fire component is always optional in the baseline.
		N ₂ O	Excluded	The project does not consider burning soil or biomass under the baseline scenario. According to the E-BPB module, the inclusion of the fire component is always optional in the baseline.
		Other	Excluded	No other GHG gases are considered in this project activity
	Burning of fossil fuels	CO ₂	Excluded	Omitted to be conservative
		CH ₄	Excluded	Omitted to be conservative

Source		Gas	Included?	Justification/explanation
Project		N ₂ O	Excluded	Omitted to be conservative
		Other	Excluded	No other GHG gases are considered in this project activity
	Oxidation of drained peat	CO ₂	Included	Considered under carbon pools
		CH ₄	Included	Required unless <i>de minimis</i> or conservatively omitted
		N ₂ O	Excluded	Excluded, as per the applicability condition in module BL-PEAT
		Other	Excluded	No other GHG gases are considered in this project activity
	Burning of peat or biomass	CO ₂	Excluded	The project did not consider the burning of soil or biomass under the baseline scenario. Thus, it was not considered under the project scenario.
		CH ₄	Excluded	The project did not consider the burning of soil or biomass under the baseline scenario. Thus, it was not considered under the project scenario.
		N ₂ O	Excluded	The project did not consider the burning of soil or biomass under the baseline scenario. Thus, it was not considered under the project scenario.
		Other	Excluded	No other GHG gases are considered in this project activity
	Burning of fossil fuels	CO ₂	Excluded	According to the methodology, potential emissions are negligible in Conservation of Intact Wetland (CIW) activities.
		CH ₄	Excluded	According to the methodology, potential emissions are negligible in CIW activities.
		N ₂ O	Excluded	According to the methodology, potential emissions are negligible in CIW activities.
		Other	Excluded	No other GHG gases are considered in this project activity

3.4 Baseline Scenario

The most plausible baseline scenario was determined using the VT0001 Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forest and Other Land Use (AFOLU) Project Activities, Version 3.0. This tool is applicable to this project as the selection of the baseline scenario can be made from a stepwise approach consistent with determining the additionality of project activities. The tool was designed for AFOLU project activities and can be used under methodology VM0007.

The methodological development for the selection of the project baseline was addressed as follows:

- 1) Identification of alternative land use scenarios to the Project AFOLU activity: Alternative land use scenarios were identified with respect to the proposed project activities, and which could be defined as a baseline scenario. The rationale for the most likely land-use scenarios in the project area is presented below.
- 2) identification of the barriers that impede the implementation of the proposed project activity which, in general terms, included financial barriers, institutional barriers, and technical barriers.
- 3) show that the identified barriers do not prevent the implementation of at least one of the alternative land-use scenarios (except for the proposed project activity): within the alternative land-use scenarios, illegal mangrove logging for charcoal production and agriculture/farmland expansion by small-scale farmers are the most common land uses in the area and are found in most rural areas. However, agriculture/farmland expansion by small-scale farmers is the alternative land use that faces the least significant identified barriers (see Table 13)

3.5 Additionality

The additionality of the project's REDD+ activity was assessed using VERRA additionality tools, namely VT0001 (AFOLU projects), v3.0, and VMD0052 (ADD-AM; tidal wetland conservation and restoration), v1.0. The VT0001 and VMD0052 have the following procedures for demonstrating additionality:

VT0001

- Step 1: identification of land-use scenarios alternative to those proposed by the AFOLU project activity;
- Step 2: investment analysis to determine that the proposed project activity is not the most economically or financially attractive of the identified land-use scenarios;
- Step 3: barriers analysis; and
- Step 4: common practice analysis.

Step 1: Identification of Alternative Land-Use Scenarios to the Proposed VCS AFOLU Project Activity

With this step, alternative land-use scenarios were identified for the proposed project. The baseline scenario was identified through the following sub-steps:

Sub-step 1a: Identify Credible Alternative Land-Use Scenarios to the Proposed VCS AFOLU Project Activity

Up to 50% of deforestation in Indonesia since 1980 has taken place in Kalimantan.⁴⁹ The main causes of deforestation in Kalimantan have, historically, been due to previously legal logging operations, land clearing by logging and fires for agriculture (both for small-scale farming and plantations), and uncontrolled forest fires, as in 1982–83, 1987, 1994, and 1997.⁵⁰ During the 1980s, when high incomes could be gained from timber trading, the Padang Tikar Landscape was renowned as an area where timber harvesting was carried out. Industrial-scale extraction in Kalimantan began in the early 1970s, which resulted in the loss of over 30% of the original forest, a rate of disappearance higher than most in the tropics. The total wood harvested from the region in the period 1980–2000 was greater than that harvested from both Africa and the Amazon combined.⁵¹ The post-harvesting area was then left abandoned as a deforested area. 40 years on, the ecosystems have still yet to fully recover from the massive harvesting era, with the forest quality currently only 30% of its initial condition. Kalimantan's remaining forests are the intact forests, found mainly at higher elevations out of reach of the logging companies, and fragmented forests in the lowlands and wetlands, including agroforestry plantations, scrublands, and croplands).⁵² Thus, the Padang Tikar Landscape consists of various types of forest ecosystem, including swamp forests (e.g., coastal mangrove forests and peatlands) and inland forests (e.g., inland peatland and dryland forests), the latter of which are composed of mineral soils.

The forest condition has since been exacerbated by climate change, experiencing a prolonged dry season and heavier rain during the rainy season. The fishing area for traditional fishermen is now decreasing and becoming more limited as a result of both climate change and the destruction of mangrove habitats. This has had the effect of decreasing the income of and increasing operational costs for traditional fishermen, despite 38% of local communities depend on fishery sectors for their livelihoods. In addition to anthropogenic factors such as the Mega Rice Project (MRP), which saw fires being used to clear forested land for agriculture (later

⁴⁹ Refer to Thompson, 1999.

⁵⁰ Refer to Boehm and Siegert, 2001.

⁵¹ Refer to Ferraz et al., 2018.

⁵² Refer to Ferraz et al., 2018.

detailed below), forest fires also occurred in 1997, being aggravated as a result of the El Niño-Southern Oscillation.⁵³

The first step for proposing this VCS AFOLU project activity is to identify realistic and credible land-use scenarios which would have occurred in the PA in the absence of the proposed project activity, i.e., conservation land use with carbon credits. Alternative land-use scenarios for mangrove forests in the Padang Tikar Landscape village of Kubu Raya, in the West Kalimantan region of Indonesia, are identified as follows, according to Bentang Kalimantan:⁵⁴

- 1) Scenario 1: illegal mangrove logging for charcoal production;
- 2) Scenario 2: agriculture or farmland expansion (e.g., small-scale palm oil plantations, rice paddy fields, coconut plantations, and mixed crops), divided by two main actors:
 - I. agriculture or farmland expansion by private actors; and
 - II. agriculture or farmland expansion by small-scale farmers; and
- 3) Scenario 3: project activity without being registered under the VCS (i.e., conversion without credit financing).

Land use in Indonesia is divided into forest estates/*kawasan hutan* and APL.⁵⁵ Forest estates in Indonesia is designated with three functions: 1) conservation forest/*hutan konservasi*; 2) protection forest/*hutan lindung* (HL); and 3) production forest/*hutan produksi* (HP), limited production forest (HPT), and convertible production forest (HPK). The HL classification entails the protection of ecosystem services (e.g., maintaining soil fertility and erosion control) in that forest area, while HP designates an area reserved for generating forest products. In most cases, HL and HP are managed by companies and communities. However, under the Village Forest scheme, management rights belong solely to communities/villages, although certain rules must still be followed as the government retains ownership rights.⁵⁶

In 2017, the MoEF granted a Village Forest license to the Padang Tikar Landscape's LPHD. The Village Forest license is one of the six schemes in the MoEF's social forestry/*perhutanan sosial* program, as established in MoEF Regulation No. P. 83 of 2016. The overarching goal of the social forestry program, in addition to the goals of Presidential Regulation No. 2 of 2015 on Agrarian Reform, is to redistribute land and management rights to local communities, protect the environment, and improve rural livelihoods by increasing community-managed forest

⁵³ Refer to Boehm and Siegert, 2001.

⁵⁴ It should be noted that, while mining operations have been present in West Kalimantan since the 20th century (Myers et al., 2016), there are no mining operations currently underway in the Padang Tikar Landscape.

⁵⁵ Refer to Santika et al., 2017.

⁵⁶ Refer to Rakatama and Pandit, 2020.

estates from 1.4 million ha to 12.7 million ha by 2019.⁵⁷ ‘Village Forest’/*Hutan Desa* has been defined as a village-managed state forest with the aim of improving that village’s prosperity and welfare.⁵⁸ To elaborate, Regulation No. P. 49 (2008) outlines that village institutions are responsible for reducing deforestation and forest degradation while safeguarding community livelihoods. Regulation P. 89, which later replaces Regulation No. P.49, echoes the same sentiment: that villages as a legal entity have the authority to regulate and manage community interest. The Village Forest scheme provides the community with a relatively long-term and extendable lease of 35 years, set to be reviewed every five years (Regulation No. P. 37 of 2007), wherein village cooperatives and village-owned enterprises have access to certain permitted areas of the forest, as detailed below:⁵⁹

- 1) HP with no licenses;
- 2) HL with no licenses;
- 3) HL managed by state-owned enterprises; and
- 4) certain designated areas within the KPH.

Permitted activities within the Village Forest area include:

- 1) utilization of the area, ecosystem services, and harvest of non-timber forest products (applies to HL area); and
- 2) utilization of the area, ecosystem services, timber products, non-timber forest products (excluding palm oil), harvest of non-timber forest products, and cultivation of timber crops (applies only to HP area).

Figure 10 shows that the majority of the Padang Tikar Landscape is HL (69,299 ha). However, the second clause of the permitted activities also applies to the Landscape, as a proportion of it is HL (1,985 ha of HPT and 5,086 ha of HPK).⁶⁰ Table 12 shows details of each Village Forest in the Padang Tikar Landscape.

Table 12: Details of the Village Forest in the Padang Tikar Landscape

⁵⁷ Refer to BAPPENAS, 2014; Indonesia MoEF, 2015; Indonesia MoEF, 2016 in Putraditama, 2018.

⁵⁸ Refer to MoEF, n.d.; Rakatama and Pandit, 2020.

⁵⁹ Refer to Indonesia MoEF 2016a in Putraditama, 2018.

⁶⁰ Refer to Village Forest of Padang Tikar: “Enhancing a Fair and Sustainable Forest Management towards Community Prosperity.”

No.	Village	Forest function and status (ha)			
		Protected forest	Limited forest	Convertible forest	Total area (ha)
1	Ambarawa	2,935		1,840	4,775
2	Batu Ampar	31,550	1,590		33,140
3	Medan Mas	1,845			1,845
4	Nipah Panjang	6,830		110	6,940
5	Padang Tikar I	190			190
6	Sungai Besar	2,900			2,900
7	Sungai Jawi	3,290		95	3,385
8	Tanjung Harapan	14,315	395	2,735	17,445
9	Tasik Malaya	2,850			2,850
10	Teluk Nibung	2,595		305	2,900
Total		69,300	1,985	5,085	76,370

HP are the only state forests where timber harvesting is permitted. However, within a Village Forest, there are annual limits to the amount of timber and non-timber products allowed to be collected per area. The utilization of this timber is governed under Law No. 41 of 1999, along with several other MoEF regulations, such as Minister Regulations on logging procedures and work plans (e.g., No. 62 of 2008, No. 65 of 2009, and No.60 of 2011) and the administration of harvested timber from natural (e.g., No. 41 of 2014) and planted forests (e.g., No. 42 of 2014), and legal certification procedures, such as the Sustainable Forest Management Certificate/*Sertifikat Pengelolaan Hutan Produksi Lestari* or Timber Legal Certificate. The utilization of timber products and the cultivation of timber crops in this sense may also refer to fibers such as pulp, paper, and/or chipwood and wood scrap.

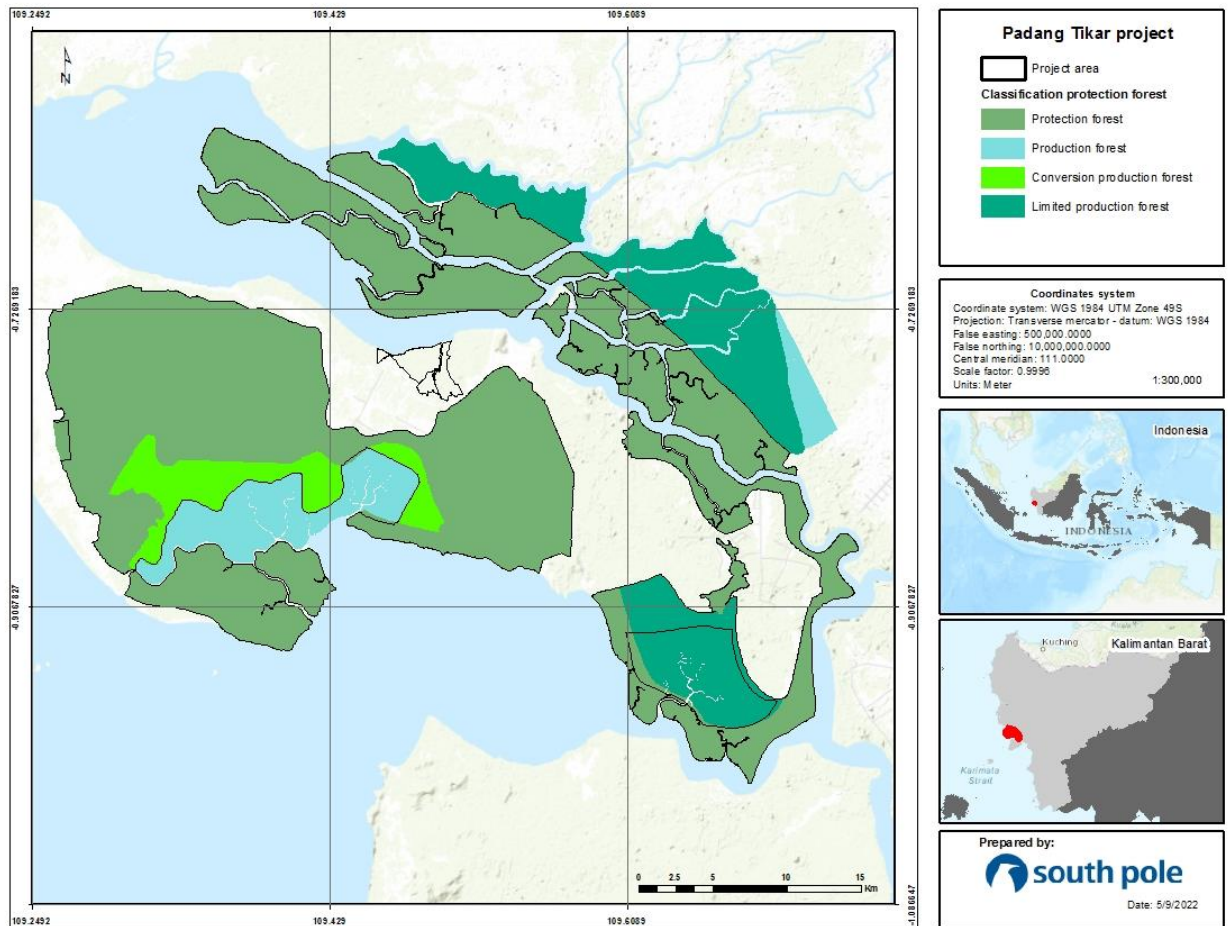


Figure 10: Village Forest of Padang Tikar Landscape

In Indonesia, the main causes of deforestation have been identified as commercial logging, land clearing for cultivation/settlement, fuelwood collection, coal mining, plantations, and fires.⁶¹ While each of the 10 villages in the proposed Padang Tikar PA have various factors contributing to deforestation associated with them, the following section provides a brief description of each of the identified alternate land-use scenarios pertinent to the villages as a whole. This is done with consideration of the relevant national or sectoral policies and circumstances, the historical uses of the land, and the practices and economic trends of the communities settled in the Padang Tikar Landscape. The following aspects may be considered: the time period; the geographical scope/boundaries; the species impacted (i.e., by type of flora and fauna); and the main products resulting from those land uses (e.g., numbers of producers and quantity of yield).

Scenario 1: Illegal Mangrove Logging for Charcoal Production

⁶¹ Refer to Thompson, 1999.

Mangroves, found predominantly in the coastal areas of the Padang Tikar Landscape, have been logged for charcoal production as, according to the local community, mangrove wood produces high-quality charcoal able to yield a high price comparative to other types of charcoal. Due to the high incomes that could previously be gained from timber trading, there was a trend of many members of the community going on to become loggers instead of progressing into education. Limited employability skills among loggers and limited access to sustainable livelihood options prevents them from diversifying to other occupational activities. Due to the extent of commercial logging in the lowlands, it is predicted that remaining forested areas will continue to be logged unless conservation plans are put in place to preserve this area.⁶²

Scenario 2: Agriculture or Farmland Expansion (e.g., Small-Scale Palm Oil Plantations, Rice Paddy Fields, Coconut Plantations, and Mixed Crops,)

Inland forests in the Padang Tikar Landscape have been cleared, either through logging or the use of fires, to make space for agriculture. As seen in **Figure B**, a large portion of the cleared land is used for settlement and cultivation. While most of the agricultural land use in the Padang Tikar Landscape is for growing rice (e.g., rice paddy fields), with almost 88% of agricultural land in West Kalimantan being used for rice growing,⁶³ other crops grown include coconut, copra, rubber, bananas, and rambutan.⁶⁴ There is also a history of growing cassava, corn, and coffee in West Kalimantan, so it is possible that these are also grown in the Padang Tikar Landscape.⁶⁵ Moreover, forested areas have also been cleared for oil palm plantations. In Bawal, West Kalimantan, mangroves have been cleared for oil palm plantations and other land uses.⁶⁶ There are a total of roughly 50 oil palm companies (i.e., private actors) present in West Kalimantan, and a total of 4,681,432 ha of land controlled by oil palm concessions.⁶⁷ The provincial government of West Kalimantan has an aim to reach a total palm oil plantation area of 4.5 million ha by 2025. Despite regulations and efforts to encourage certain areas to deviate from palm oil plantations, oil palm still plays a significant role in Indonesia's economic growth.

However, in PAs which hold Village Forest permits, it is very difficult for oil palm companies to get the license to cultivate palm oil. Changing the current land use status of either HL or HP to APL, thereby allowing these areas to be used for agricultural purposes such as palm oil, would require a protracted process at the bureaucracy level. Meanwhile, small-scale farmers have no barriers to changing land use to, for instance, palm oil or other agricultural commodities, due to

⁶² Refer to Ferraz et al., 2018.

⁶³ Refer to Myers et al., 2016.

⁶⁴ Refer to Samdhana Annual Report, 2015.

⁶⁵ Refer to Embassy of Indonesia, 1982.

⁶⁶ Refer to Quevedo et al., 2020.

⁶⁷ Refer to IDH, n.d.

their lack of awareness and pressure for economic gain. Small-scale farmers played a big role in the land-use change within the Padang Tikar Landscape. The shift from water- and sea-based economies to a land-based economy creates pressure on a Landscape that contains high biodiversity value. Moreover, with the high demand for palm oil and the existence of palm oil mills in the area, it is not unprecedented for communities to cultivate palm oil informally.

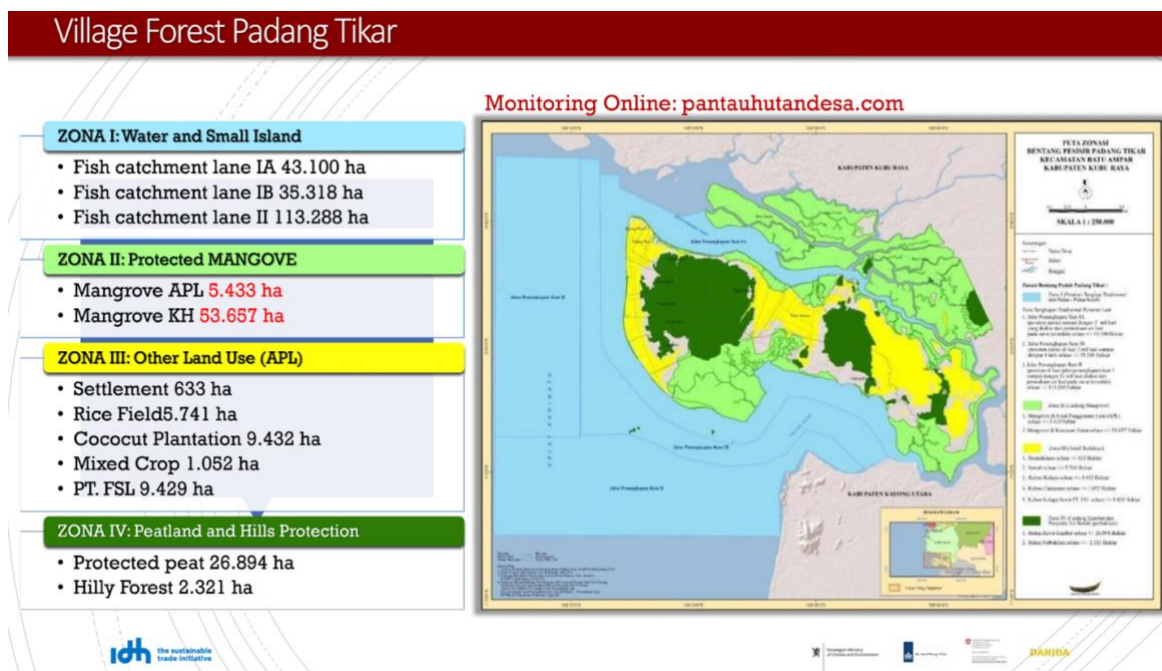


Figure 11: Land use types in Padang Tikar Landscape⁶⁸

Scenario 3: Project Activity Without Being Registered Under the VCS (i.e., Conversion Without Credit Financing)

The last alternative land-use scenario is the conservation of both inland and mangrove forests without the involvement of carbon credits, e.g., via carbon credit project development and financing. However, conservation measures have historically been met with noncompliance, as attempts to enforce it have proven ineffective. For instance, the authorization of local district governments to grant community forestry licenses so as to increase uptake of the scheme (Decree No. 31/Kpts-II in 2001) was met with the MoEF requesting re-evaluation of issued community forestry licenses (Regulation No. 722/Menhut-V) the following year, due to misuse of the concession for logging. Additionally, the effective conservation impact of the MoEF's social forestry program has also been contested. Despite the social forestry program, deforestation has continued, as alternative, non-timber harvesting activities seem insufficient in fulfilling the daily needs of local communities.⁶⁹ Previous studies in Indonesia have

⁶⁸ Available at: IDH Landscape program, Sustainable Mangrove Business Model in West Kalimantan, 2022

⁶⁹ Available at: Proposal for Social Forestry Permit, 2014

demonstrated high community awareness and understanding of protecting ecosystem services, along with awareness that renewal of a Village Forest license is contingent on community performance in conserving the forest area.⁷⁰ Despite this, there is limited interest in conservation due to the lack of direct economic incentives, especially if exploiting forest products is more profitable or able to satisfy an urgent need for income.⁷¹ From 2014 data, around 7.14% of people in the Padang Tikar Landscape were living at or below the poverty line.⁷² Considering this, conservation without economic incentives is unlikely to happen.

With these alternative land-use scenarios in mind, the generation and selling of carbon credits from forest and mangrove protection and restoration through leveraging ecosystem services, as allowed by the Village Forest permit, can be one way of meeting the goals of the Village Forest permit and avoiding deforestation.

Outcome of Sub-step 1a:

Thus, the following scenarios are credible alternative land-use scenarios that may have occurred on the land within the AFOLU project activity boundary:

- 1) illegal mangrove logging for charcoal production;
- 2) agriculture/farmland expansion by private actors and/or small-scale farmers; and
- 3) conservation without carbon credit financing.

Sub-step 1b: Consistency of Credible Alternative Land-Use Scenarios with Enforced Mandatory Applicable Laws and Regulations

As mentioned above, there are three possible alternative land-use scenarios in the Padang Tikar PA. Scenario 3, or conservation without carbon credit financing, is the only land-use scenario that would actively comply with the national mandatory legislation and regulations for the Padang Tikar Landscape.

This leaves two remaining scenarios: scenario 1, or illegal mangrove logging for charcoal production, and scenario 2, which is divided into scenario 2.1 (agriculture/farmland expansion by private actors) and scenario 2.2 (agriculture/farmland expansion by small-scale farmers). As explained in sub-step 1a of this section, scenario 2.1 is unlikely to happen due to political barriers. This leaves us with scenarios 1 and 2.2 as non-compliant with legislation and regulations, both of which are widespread as a result of the systematic lack of enforcement of applicable laws and regulations (see scenario 3 in sub-step 1a of this section). One such rule or provision of forest management rights specified by the Padang Tikar Village Forest permit

⁷⁰ Refer to Quevedo et al., 2022; Ribot, 2004 in Putraditama, 2018.

⁷¹ Refer to Quevedo et al., 2022; Ribot, 2004 in Putraditama, 2018.

⁷² Available at: Proposal for Social Forestry Permit, 2014.

prohibits the Village Forest from being used for purposes other than those mentioned in the RPHD (i.e., conservation), along with the collection and utilization of timber forest resources and environmental services (Regulation No. P. 83 of 2016 and Regulation No. P. 37 of 2007). As agriculture would require forests to be cleared, agriculture would be non-compliant with the rules of the permit. Specifically, the permit also prohibits revegetating the Village Forest area with palm oil species. Under this rule, the use of the Padang Tikar Landscape for palm oil plantations would also have been illegal from the date the permit was issued in 2017. However, as the rules against agricultural expansion and palm oil plantations are not enforced, evident from how these activities continue to be carried out in the Padang Tikar Landscape, these land uses can remain on the list of possible land uses, despite their non-compliance with current laws and regulations.⁷³

Outcome of Sub-step 1b:

All three scenarios above (except for scenario 2.1) are plausible alternative land-use scenarios to the VCS AFOLU project activity, either being compliant with mandatory legislation and regulations or resulting from a systemic lack of enforcement of applicable laws and regulations.

Sub-step 1c: Selection of the Baseline Scenario

The baseline scenario will be chosen based on the scenarios presented.

Step 2: Investment Analysis

Not applicable for this project.

Step 3: Barrier Analysis***Sub-step 3a: Identify Barriers That Would Prevent the Implementation of the Type of Proposed Project Activity*****Financial barriers**

While LPHDs have a mandate to manage the Village Forest, there is no financial support for LPHDs in implementing activities necessary for maintaining the Village Forest permit. Knowing this, the government requires the presence of an accompanying facilitator to support the implementation of any established Village Forest permit on the ground. This role is often filled by a local NGO. In this project, Bentang Kalimantan is the local NGO that will support the LPHD with both the implementation of the Village Forest permit mandate and the proposed carbon asset development activity as a responsible entity. The LPHD is required to develop a management plan for the implementation of the permit, however, without support from local NGOs – which also depend on the support from donor agencies – the implementation of the management plan would be difficult.

⁷³ Refer to Mongabay, 2020.

The conservation activity in the Padang Tikar Landscape is highly dependent on donor support. Previous conservation efforts have been supported by the IDH and other donor agencies, such as the United Nations Development Programme, the Indonesia Climate Change Trust Fund, and Yayasan Belantara. Considering the fact that forest conservation at scale did not occur in the Padang Tikar Landscape prior to the involvement of donors and local NGOs, it is unlikely that villagers would continue to preserve the forests without financial support to provide income security for their conservation efforts.⁷⁴ Contributing to this is that conservation in the area is seldom sufficiently carried out without carbon credit financing (see step 4: common practice analysis). In being highly dependent on donor support, the current financial model is limiting the long term sustainability of project activities. As such, carbon credits could be the bridge to allowing Village Forests to strengthen their capacity for implementing project activities in the long run.

Institutional barriers

The bureaucracies and complexities surrounding the social forestry program also prevent the community from fully understanding the nature and merits of such forest conservation programs. While Padang Tikar already has a Village Forest permit, the process of obtaining this is fairly lengthy and can take time to complete. When pursuing a permit, communities have to decide whether to complete an application to manage the forest as a single landscape overseen by one village, or to split the application up by each village. The application would also have to include a Village Forest proposal in the social forestry scheme and revisit the regulations to become one.⁷⁵ As an indicator, only around one out of approximately 3,000 villages in Indonesia's peatlands have received a permit to manage its forests under the social forestry program⁷⁶. Similarly, the complex regulations surrounding social forestry programs can be a cause of confusion, as studies have indicated that those signing up for social forestry schemes often do so with limited understanding of what the schemes entail.⁷⁷

Moreover, village communities are often unaware of the rights and responsibilities they have under the Village Forest permit.⁷⁸ The role of local NGOs in assisting LPHDs has become crucial to the implementation of Village Forest permits. In the 'Theory of Change' for this project, Bentang Kalimantan aims to increase the leadership of LPHDs in implementing, monitoring, and evaluating the proposed project activities, as current baselines indicate that the LPHD has little experience in and capacity for implementing the Village Forest mandate. Without

⁷⁴ Refer to Samdhana, 2015.

⁷⁵ Refer to Samdhana, 2015.

⁷⁶ Refer to Mongabay, 2018.

⁷⁷ Refer to Rakatama and Pandit, 2020.

⁷⁸ Refer to Rakatama and Pandit, 2020.

assistance, the Village Forest permit they receive, evaluated every five years, may be revoked by the government if the results show that the LPHD failed to implement their responsibility under the permit.

Technological barriers

As the proposed project activities also require proposing alternative livelihoods to the community, their capacity prior to the project was only focused on the producers' side, often lacking a pasca-producing stage to add value to their production. Not only that, but the community also has a lack of knowledge on how to manage the land, with their prior experience focused more so on managing water- and sea-based product. Without assistance from a local NGO, the community may not be aware of the potential of their land, or how to improve the water and sea products still allowed by current laws and regulation. Technological barriers are also very closely linked to financial barriers. If there is no financing to overcome the technological barriers, the implementation of proposed project activities would not be possible. For example, the community needs support with pasca-producing activities, such as on the packaging and marketing side, to add to the value of the products. It would otherwise be very difficult for the community to overcome these barriers without this additional support.

Outcome of Sub-step 3a:

In conclusion, there are three main barriers that prevent the implementation of project activities:

- 1) financial barriers, specifically for the implementation of project activities related to climate and biodiversity;
- 2) institutional barriers, especially regarding the capacity of the LPHD, affecting the enforcement of laws in the permit area; and
- 3) technological barriers, specifically in terms of introducing alternative livelihoods for the community.

Sub-step 3b: Show That the Identified Barriers Would not Prevent the Implementation of at Least one of the Alternative Land-use Scenarios (Excluding the Proposed Project Activity)

Based on Table 13 below, the alternative land-use scenario that faces the least significant barriers is scenario 2.2, or agriculture/farmland expansion by small-scale farmers, e.g., for small-scale palm oil plantations, rice paddy fields, coconut plantations, and mixed crops.

Table 13: Alternative land-use scenarios and identified barriers

Alternative land-use scenarios	Description of barriers
Scenario 1: Illegal mangrove logging for charcoal production	The most obvious barriers to illegal logging are the ecological and technological barriers, as logging can only

Alternative land-use scenarios	Description of barriers
	be carried out in accessible areas. The topographic variation of well-drained lowland dipterocarp forest intermixed with swamp forest in West Kalimantan served as a barrier to illegal logging, as heavy machinery is unable to pass through these areas. This restricts logging to small-scale activities. Prevailing practices may also contribute to illegal logging, as it has been found that, despite social forestry designations, locals often continue to practice activities that were previously practiced in the area. Additionally, it is assumed that the same subset of people that engage in illegal logging would be involved in other livelihood opportunities related to agricultural/farmland expansion (i.e., scenario 2.2), which also involve the unsustainable use of the forested land. If so, these other livelihood opportunities would act as a barrier to illegal logging activity, as they would compete with scenario 1. It is noted that this is a barrier resulting from prevailing practices and would also be applicable to scenario 2.2. Finally, enforced land tenure agreements within the community may restrict the areas in which the individual logger may be able log. While the land ownership agreements and rights may be informal, the barrier of land tenure rights would nevertheless be applicable in this scenario. Moreover, the land function change from HL to HP required to allow timber harvesting would also be difficult to implement, as the Padang Tikar area is recognized as an important landscape for mangroves.
Scenario 2.2: agriculture/farmland expansion by small-scale farmers, e.g., for small-scale palm oil plantations, rice paddy fields, coconut plantations, and mixed crops	There are insignificant barriers to this land-use scenario. In fact, there is some government support for this activity. • The Padang Tikar community grows rice as subsistence farming, as rice is a staple food for Indonesia. In 1995, under Presidential Decree No. 82: Development of One Million Hectares of Peatland for Food Crop Production in the Province of Central Kalimantan, Peat Reclamation (also known as the Peat Area Project/<i>Proyek Lahan</i>

Alternative land-use scenarios	Description of barriers
	<i>Gambut</i>), the government initiated the MRP, a large-scale peatland conversion program which saw large-scale land clearing by fire and the creation of more than 4,000 km of drainage/irrigation channels within MRP-designated areas between 1996–1998, the latter of which (drainage) also contributed to fires.⁷⁹ The MRP identified some districts in Kalimantan as suitable for rice cultivation. While local communities in Central Kalimantan traditionally cultivated rice, this was done at a small-scale and on shallow peatland, boasting limited environmental impact.⁸⁰ However, for the MRP, 165,000 ha of wetlands in Central Kalimantan’s Pulang Pisau and Kapuas districts were identified as suitable for rice cultivation, while in Rimba Raya in Central Kalimantan, forested land has been cleared for industrial-scale rice production.⁸¹ Against this background and the prevailing practice in the region, it is likely Padang Tikar was affected by the same or similar initiatives as its neighbors in expanding its farmlands. Moreover, the Indonesian government has launched a new food estate program which will target peatland areas to be managed for rice fields and other food crops, although the current focus is still on Central Kalimantan. • In terms of coconut production, coconut has been grown and consumed in Indonesia for centuries. There is also increased global demand for coconut products, which will increase the opportunity for farmers to produce this commodity.⁸²

⁷⁹ Refer to Boehm and Siegert, 2001.

⁸⁰ Refer to Boehm and Siegert, 2001.

⁸¹ Refer to Mongabay, 2020a.

⁸² Refer to Mongabay, 2020b.

Alternative land-use scenarios	Description of barriers
	• Small-scale farmers have little to no awareness of the regulation barriers. Furthermore, enforcement of the law is weak, and no sanctions from the government are in place to overcome this issue. • There was a request from the community during the LSC to support agricultural commodities from project activities in the Landscape to improve their economy. This is a sign that the local community is aware of the potential agriculture may pose for their livelihoods.

Step 4: Common Practice Analysis

In Indonesia, only several conservation areas have been designated as either HL or HP or issued with Village Forest permits since the start of the social forestry program. Village Forest areas are not a common practice. Few efforts have been made to develop ecologically sustainable livelihood alternatives, and no other forest conservation activities of a similar scale to those proposed by the Padang Tikar Landscape project have been developed, implemented, or are currently underway in the West Kalimantan region. The project seems to be one of the few regional conservation activities that combines multiple ecosystems, including dryland forest, mangrove forest, and swamp forest, under its protection, with most conservation activities found focusing on just one type of ecosystem. From this, it seems that activities that encompass a variety of wetland ecosystems are not a common practice in the region. Additionally, the Padang Tikar Landscape Village Forest license also covers the greatest area in Indonesia, encompassing around 75,000 ha of mangrove and peat forest.⁸³

In conclusion, the proposed project activity is not a common practice in the West Kalimantan region and the country, and, therefore, the proposed VCS AFOLU project activity is additional.

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- Step 1: regulatory surplus
- Step 2: positive list

Step 1: Regulatory Surplus

The proposed project activities sit within the legal and regulatory frameworks for HL and HP (Law No. 41 in 1999), social forestry (Regulation No. P.83 of 2016), and Agrarian Reform Presidential Regulation No. 2 of 2015 for managing Indonesia's

⁸³ Available at: <https://www.idhsustainabletrade.com/news/indonesia-landscape-alternative-livelihoods/>.

forests through redistributing land and management rights to local communities, protecting the environment, and improving rural livelihoods. The generation and selling of carbon credits from forest and mangrove protection and restoration is a form of utilizing the area's ecosystem services without harvesting timber products, in alignment with the Village Forest permit's permitted activities (Regulation No. P.83 of 2016 and Law No. 41 in 1999). Crab and honey production activities also serve to improve rural livelihoods without exploiting timber products and deforestation (Agrarian Reform Presidential Regulation No. 2 of 2015).

The proposed VCS AFOLU project activity is not bound by current legal and regulatory requirements, nor is the PA involved in any mandatory carbon market-related schemes. While the social forestry program provides a framework for conservation, there is no clear guidance or finance scheme to support villages with achieving the aims of the Village Forest permit. With this, the proposed project activities would be additional to and supporting Regulations No. P. 49 and No. P. 89, which outline the responsibility of village institutions and their relevant authorities to manage the forests in line with conservation regulations (i.e., social forestry) and through providing training in sustainable livelihood options. Additionally, it has been found that, while Village Forests are useful for avoiding deforestation overall, Village Forests that fall under the categories of Watershed Protection Forests or HPT, or those areas of the proposed project activity where most Village Forest permits have been granted, have less avoided deforestation.⁸⁴ Supporting this is evidence that local forest people are more likely to engage in conservation and participate in accordance with social forestry schemes if they see a value in doing so. Thus, the proposed project activities suggest livelihood activities and financing conservation through carbon credits as forest conservation efforts, seeking to fill the gap between incentivizing and facilitating conservation. However, previously, the authorization of local district governments to grant community forest licenses (Decree No. 31/Kpts-II in 2001) led to the MoEF requesting the re-evaluation of issued community forest licenses in 2002 due to misuse of the concession for logging (Regulation No. 722/Menhut-V). Furthermore, alternative, non-timber harvesting activities without economic incentives are not an attractive enough option for local communities in lieu of needing income to fulfil daily needs.

Step 2: Positive List

In reference to Table 9, the proposed project activity fulfils the applicability conditions of the VM0007 tool and is subsequently in compliance with the positive list.

3.6 Methodology Deviations

This project activity does not apply any methodology deviation.

⁸⁴ Refer to Santika et al., 2017.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Describe the procedure for quantification of baseline emissions and/or removals in accordance with the applied methodology. Include all relevant equations, and explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values).

4.2 Project Emissions

Describe the procedure for quantification of project emissions and/or removals in accordance with the applied methodology. Include all relevant equations, and explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values).

4.3 Leakage

Describe the procedure for quantification of leakage emissions in accordance with the applied methodology. Include all relevant equations, and explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values).

4.4 Net GHG Emission Reductions and Removals

Describe the procedure for quantification of net GHG emission reductions and removals. Include all relevant equations. For AFOLU projects, include equations for the quantification of net change in carbon stocks.

Provide the ex-ante calculation (estimate) of baseline emissions/removals, project emissions/removals, leakage emissions and net GHG emission reductions and removals in the table below.

For data and parameters monitored, use estimates. Document how each equation is applied, in a manner that enables the reader to reproduce the calculation. Provide example calculations for all key equations, to allow the reader to reproduce the calculation of estimated net GHG emission reductions or removals.

Year	Estimated baseline	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage	Estimated net GHG emission
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	emissions or removals (tCO ₂ e)		emissions (tCO ₂ e)	reductions or removals (tCO ₂ e)
Year A				
Year B				
Year C				
Year...				
Total				

5 MONITORING

5.1 Data and Parameters Available at Validation

Complete the table below for all data and parameters that are determined or available at validation, and remain fixed throughout the project crediting period (copy the table as necessary for each data/parameter). Data and parameters monitored during the operation of the project are included in Section 5.2 (Data and Parameters Monitored) below.

Data / Parameter	
Data unit	Indicate the unit of measure
Description	Provide a brief description of the data/parameter
Source of data	Indicate the source(s) of data
Value applied	Provide the value applied
Justification of choice of data or description of measurement methods and procedures applied	Justify the choice of data source, providing references where applicable. Where values are based on measurement, include a description of the measurement methods and procedures applied (e.g., what standards or protocols have been followed), indicate the responsible person/entity that undertook the measurement, the date of the measurement and the measurement results. More detailed information may be provided in an appendix.
Purpose of Data	Indicate one of the following: - Determination of baseline scenario (AFOLU projects only) - Calculation of baseline emissions - Calculation of project emissions - Calculation of leakage

Comments	Provide any additional comments
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5.2 Data and Parameters Monitored

Complete the table below for all data and parameters that will be monitored during the project crediting period (copy the table as necessary for each data/parameter). Data and parameters determined or available at validation are included in Section 5.1 (Data and Parameters Available at Validation) above.

Data / Parameter	
Data unit	Indicate the unit of measure
Description	Provide a brief description of the data/parameter
Source of data	Indicate the source(s) of data
Description of measurement methods and procedures to be applied	Specify the measurement methods and procedures, any standards or protocols to be followed, and the person/entity responsible for the measurement. Include any relevant information regarding the accuracy of the measurements (e.g., accuracy associated with meter equipment or laboratory tests).
Frequency of monitoring/recording	Specify measurement and recording frequency
Value applied	Provide an estimated value for the data/parameter
Monitoring equipment	Identify equipment used to monitor the data/parameter including type, accuracy class, and serial number of equipment, as appropriate.
QA/QC procedures to be applied	Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures where applicable.
Purpose of data	Indicate one of the following: - Calculation of baseline emissions - Calculation of project emissions - Calculation of leakage
Calculation method	Where relevant, provide the calculation method, including any equations, used to establish the data/parameter.
Comments	Provide any additional comments

5.3 Monitoring Plan

Describe the process and schedule for obtaining, recording, compiling and analyzing the monitored data and parameters set out in Section 5.2 (Data and Parameters Monitored) above. Include details on the following:

- *The methods for measuring, recording, storing, aggregating, collating and reporting data and parameters. Where relevant, include the procedures for calibrating monitoring equipment.*
- *The organizational structure, responsibilities and competencies of the personnel that will be carrying out monitoring activities.*
- *The policies for oversight and accountability of monitoring activities.*
- *The procedures for internal auditing and QA/QC.*
- *The procedures for handling non-conformances with the validated monitoring plan.*
- *Any sampling approaches used, including target precision levels, sample sizes, sample site locations, stratification, frequency of measurement and QA/QC procedures.*

Where appropriate, include line diagrams to display the GHG data collection and management system.

APPENDIX

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.